The one-sided CHSH measure under local filtering: exact supremum, violation criterion and a least-eigenvalue formula

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Abstract

Ducuara, Susa and Skrzypczyk (Phys. Rev. A **110**, 022435, 2024) defined the one-sided hidden CHSH measures HB_A and HB_B of a two-qubit state, the largest normalized CHSH value obtainable when only one party applies an invertible local filter (their Eq. (9)), and noted that no closed formula was known. We solve this optimization for every two-qubit state. After whitening the filtering party's marginal, the supremum equals the maximum of one explicit quadratic function over a compact set of pairs of vectors in $\mathbb{R}^3$; it is attained by an invertible filter whenever it exceeds the local bound, and in general may be only a supremum. Writing the whitened data as an other-party Bloch vector c and correlation matrix D, put $A = DD^{\mathsf{T}}$, $K = A - \|c\|^2 I$, $C = [Dc]_\times$ and $\alpha = \operatorname{tr} A - \|c\|^2$. One-sided violation is possible only if $\alpha > 1$, and throughout that region the squared supremum is $\operatorname{tr} A - \lambda_{\min}(A + CK^{-1}C)$, the least eigenvalue of an explicit symmetric 3×3 matrix, with an explicit optimal filter. Hence violation occurs exactly when $\alpha > 1$ and $A + CK^{-1}C - (\operatorname{tr} A - 1)I$ is not positive semidefinite, a test by three polynomial signs. For $\alpha \le 1$ the value is at most one and is the largest of at most four algebraic candidates: values at the admissible roots of a cubic pencil and a boundary radius determined by a specified root of a secular equation of degree at most six. We add a rational formula for states of rank at most two, the exact one-sided value $\max\{1, 2p/(2-p)\}$ for the quasi-distillable family of the 2024 paper, and exact success-probability trade-offs. Lean 4 with Mathlib proves, for every positive semidefinite trace-one complex 4×4 matrix, arbitrary invertible complex filters on either party and arbitrary traceless Hermitian involutive qubit observables, the supremum, the violation criterion, the least-eigenvalue formula and invertible attainment; the secular-root boundary algorithm and the further results have conventional proofs and independent checks.

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1 Introduction

1.1 The question in plain terms

Two parties, Alice and Bob, share a two-qubit state ρ and run a CHSH Bell test. A state that does not violate the CHSH inequality may still do so after *local filtering*: each party applies a two-outcome measurement, and the parties keep the runs in which a designated outcome occurred. Filtering by both parties is completely understood: the optimal filters bring the state to a Bell-diagonal normal form, and the resulting "hidden" CHSH value has a closed formula [4, 5, 1]. Ducuara, Susa and Skrzypczyk [1] asked what happens when only *one* party may filter, a situation natural in semi-device-independent settings. They defined

$$HB_W(\rho) = \max_{f_W} B(\rho'_{F_W}), \qquad W \in \{A, B\}, \tag{1}$$

their Eq. (9), where B is the CHSH value normalized to local bound 1 and Tsirelson bound $\sqrt{2}$, $f_W \in \mathrm{GL}(2, \mathbb{C})$ acts on the party W only, and ρ'_{F_W} is the normalized filtered state. They observed that "unlike the standard hidden correlations, ... there are currently no closed formulas for the one-sided versions", and listed closed formulas for these measures among their open questions. Below we write B^*_A for HB_A.

1.2 The answer in plain terms

The filtering party's marginal can first be made maximally mixed ("whitened") without changing the set of reachable states. After that, a filter is described by a vector u in the Bloch ball, and the optimal CHSH value of the filtered state is a ratio of quadratic expressions. A change of variables (Section 4) turns the whole problem, including the choice of the two measured directions, into the maximum of a single quadratic function $f(n, v)$ over pairs of vectors with $\|n\| = 1$, $n \perp v$ and $\|v\| \leq 1$. Points with $\|v\| = 1$ correspond to rank-one limits of invertible filters, which is why the "maximum" in Eq. (1) is in general only a supremum.

Physical positivity then does most of the work. A spin-flip Gram identity shows that a one-sided violation needs the purity quantity α of the whitened state to exceed one, and that for $\alpha > 1$ the maximum of f is interior and is given by completing a square: the answer is one

explicit 3×3 least eigenvalue. For $\alpha \leq 1$ no violation is possible; the exact sub-classical value is still determined, as the largest of finitely many algebraic candidates.

1.3 Main results

Let ρ have Bloch data (a, b, T) and mixed Alice marginal ($\|a\| < 1$). Whitening Alice gives canonical data (c, D) (Equation (6)); put

$$A = DD^{\mathsf{T}}, \quad s = \|c\|^2, \quad C = [Dc]_\times, \quad K = A - sI, \quad \alpha = \operatorname{tr} A - s, \quad M = A + CK^{-1}C.$$

In original coordinates $\alpha = (\|T\|_F^2 - \|b\|^2)/(1 - \|a\|^2)$ (Equation (34)). Bob's measure is obtained by exchanging the parties, that is, exchanging a and b and transposing T.

A. Exact supremum for every state. $(B_A^*)^2$ equals the maximum of the explicit quadratic f of Equation (15) over the compact domain (16); if Alice's marginal is pure, the state is a product and $B_A^* = \|b\|$, attained (Equations (20) and (7)). *Lean-checked.*

B. Least-eigenvalue formula where violation is possible. If $\alpha > 1$ then $K \succ 0$ and

$$(B_A^*)^2 = \operatorname{tr} A - \lambda_{\min}(M),$$

attained by the invertible filter of Equation (36) (Theorem 3). *Lean-checked.* The same holds when $K + iC \succ 0$ (conventional).

C. Exact violation criterion. $B_A^* > 1$ if and only if $\alpha > 1$ and $M - (\operatorname{tr} A - 1)I \not\succeq 0$; a violating supremum is always attained by an invertible filter. Hidden one-sided activation additionally requires $s_1(T)^2 + s_2(T)^2 \leq 1$ for the original state. *Lean-checked.* The equivalent test by the signs of three polynomials in the Bloch data, including a version in rational steering-ellipsoid data (Theorem 10), is conventional.

D. All remaining states. For $\alpha \leq 1$, $(B_A^*)^2 \leq 1$, and for every state $(B_A^*)^2$ is the largest of the boundary radius $E = \max_{\|u\|=1} \left\|c + D^{\mathsf{T}}u\right\|^2$ and at most three values $\operatorname{tr} A - \lambda$ at admissible roots λ of a cubic pencil (Theorem 6). The finite-candidate statement is *Lean-checked*, with E represented as an attained maximum over the sphere; the explicit evaluation of E by a secular equation and the admissibility tests are conventional.

E. Further exact results (conventional, with exact and numerical checks): for $\operatorname{rank}\rho \leq 2$, $(B_A^*)^2 = 1 + \max(0, \|T\|_F^2 + \|a\|^2 - \|b\|^2 - 1)/(2(1 - \|a\|^2))$ (Theorem 15); $B_A^* = \sqrt{2}$ exactly for pure entangled states; a three-regime classification at $B_A^* = 1$ (Theorem 13); the exact one-sided value $(B_A^*)^2 = (B_B^*)^2 = \max\{1, 2p/(2-p)\}$ for the quasi-distillable family $p\,\psi^- + (1-p)|00\rangle\langle 00|$ of [1]; exact success-probability trade-offs for canonical and original inputs (Sections 14 and 15); and convexity, noise-scaling and stability properties (Section 16).

What is closed-form and what is algorithmic. Results B and C are closed formulas: a least eigenvalue of an explicit symmetric 3×3 matrix (a root of an explicit cubic, Corollary 11) and three polynomial sign conditions. They cover every state for which one-sided violation is possible, and every state with $\alpha > 1$ whether or not it violates. The rank-two formula and the pure-marginal value are also closed. For $\alpha \leq 1$ the value never exceeds one, and its exact value is algorithmic but finite: compute the cubic pencil roots, test their admissibility by small singular-value problems, evaluate E through the unique root above the top eigenvalue of A of a secular equation of degree at most six (or the explicit hard-case expression), and take the largest. In the Lean development the value for $\alpha \leq 1$ appears as this finite candidate set with E an attained maximum over a sphere; the secular-root construction of E is not formalized.

Formal scope. The Lean statements quantify over every positive semidefinite trace-one 4×4 complex matrix, every complex 2×2 filter with nonzero determinant on either party, normalized by the actual trace, and every quadruple of traceless Hermitian involutive qubit matrices as measurement observables; Appendix B lists the declarations and what remains conventional.

Reading guide. Section 2 records prior work and priority. Section 3 fixes notation and reduces to a whitened marginal; Section 4 proves the compact reduction (Result A). Sections 5–6 prove Result B, which is all that is needed for the violation question. Section 7 treats every state (Result D); the proof of its finite-candidate theorem is in Appendix A. Section 9 gives optimal filters, attainment and the violation criteria (Result C), and the later sections contain the further results. Appendix B describes the verification.

2 Prior work and priority

The question. The measures (1) and the normalization of B are those of Ducuara, Susa and Skrzypczyk [1]. The optimal CHSH value over projective qubit measurements is the Horodecki formula $B(\rho)^2 = s_1(T)^2 + s_2(T)^2$ [2], used throughout. The Lorentz representation of local filtering is due to Verstraete, Dehaene and De Moor [3]; the optimal bilateral filters and the Bell-diagonal normal form are due to Kent, Linden and Massar [5] and Verstraete and Wolf [4]. Bilateral hidden-nonlocality criteria include those of Pal and Ghosh [6] and Li et al. [7]; they optimize both filters and do not solve (1). Whitening one party identifies its filtering orbit with pure inputs to the qubit channel whose Choi state is the whitened state, so one-sided activation is equivalent to failure of the CHSH-breaking property for that channel; Zhang, Araiza Bravo, Lorenz and Chitambar [8] solve the unital case (recovered here at $c = 0$) and several nonunital families, including the amplitude-damping threshold, and state that no general analytical criterion was known for nonunital channels. Kumari et al. [9] relate CHSH breaking to incompatibility breaking.

Ingredients that are not new. The bound of the CHSH value by a conditional purity (negative conditional Rényi-2 entropy is necessary for violation) is due to R. Horodecki, P. Horodecki and M. Horodecki [10]. The rank-two formula uses the Ruskai–Szarek–Werner normal form of qubit channels [11]. The boundary radius is a trust-region subproblem, evaluated by the classical secular equation and hard case of Moré and Sorensen [12]. The certificate hierarchy in Section 15 uses the matrix Positivstellensatz of Hol and Scherer [13].

Omega Institute record on Milne's Conjecture 1 (note dated 3 November 2026). The Omega Institute's public repository `the-omega-institute/trureturing` records a claimed Lean proof of Conjecture 1 of Milne, Jennings, Jevtic and Rudolph [14], the bound of the CHSH value by the length c of the centre of the steering ellipsoid [15] for every two-qubit state; the dossier for its issue #16558 was committed on 14 October 2026 at 08:40:22 UTC [16]. That record touches this paper in two places. First, its proof uses a plane-purity identity and the covariance of a conditional purity under one-sided filtering; with the parties exchanged these are Equations (33) and (30) of Proposition 2, which we therefore do not claim as new. Second, Result 1 of [1] deduces the impossibility of one-sided violation for centre lengths above $1/2$ from Conjecture 1, so the claimed proof makes that criterion unconditional; for the quasi-distillable family it already gives the inaccessibility interval $0 < p \le 2/3$ (Section 13), for which no priority is claimed. The package chsh-steering-ellipsoid of this collection treats Conjecture 1 as well. Neither record, as read on 3 November 2026, contains the one-sided supremum, the exact violation criterion, the least-eigenvalue formula or the finite candidate formula, and Conjecture 1 is not used here. The Omega dossier lists bounds on the CHSH value by the full steering ellipsoid,

rather than its centre alone, as open. We have not checked the Omega Institute's Lean proof and describe it as claimed.

What is new here, as far as we found. The compact reduction, the finite spectral formula for every state, the least-eigenvalue formula with automatic feasibility in the region $\alpha > 1$, the exact one-sided violation criterion and its polynomial form, the rank-two formula for the one-sided measure, the exact one-sided values for the quasi-distillable family, the success-probability trade-offs and the Lean development. A bounded literature search did not find these results; this does not establish priority, and a specialist review remains appropriate.

3 Setting, notation and the canonical reduction

Write a two-qubit density matrix as

$$\rho = \frac{1}{4}\sum_{\mu,\nu=0}^{3} R_{\mu\nu}\sigma_\mu \otimes \sigma_\nu, \qquad R = \begin{pmatrix} 1 & b^T \\ a & T \end{pmatrix}. \tag{2}$$

We use precisely the normalization of Equations (2) and (9) of Ducuara–Susa–Skrzypczyk [1]; the expression of the optimal CHSH value through the two largest singular values $s_1(T) \geq s_2(T)$ of the correlation matrix is the Horodecki criterion [2]:

$$B(\rho) = \sqrt{s_1(T)^2 + s_2(T)^2}, \qquad B_A^*(\rho) = \sup_{F\in \mathrm{GL}(2,\mathbb{C})} B\left(\frac{(F\otimes I)\rho(F^\dagger \otimes I)}{\mathrm{tr}[(F^\dagger F \otimes I)\rho]}\right). \tag{3}$$

The CHSH local bound is 1, the Tsirelson bound is $\sqrt{2}$, and the usual unnormalized Bell expression is $2B$. Only the traceless spin observables in the paper's definition are being optimized. Consequently B_A^* can be less than 1, or even zero. It is incorrect to insert a universal floor of 1.

The paper writes a maximum over invertible filters. A supremum is necessary: the maximizing sequence can converge to a rank-one filter. All positive scalar multiples of a filter yield the same normalized state; a scalar can always make $F^\dagger F \leq I$.

If $\rho_A = (I + a\cdot\sigma)/2$ is invertible, set

$$W = (2\rho_A)^{-1/2}, \qquad \rho_c = (W\otimes I)\rho(W\otimes I), \qquad R_c = \begin{pmatrix} 1 & c^T \\ 0 & D \end{pmatrix}. \tag{4}$$

The trace of ρ_c is 1. Since W is invertible, composing filters with W gives a bijection of the unilateral filtering orbit. Thus it suffices to compute $B_A^*(\rho_c)$. For direct Bloch-data evaluation, with $r_a = \sqrt{1-\|a\|^2} > 0$,

$$c = \frac{b - T^T a}{r_a^2}, \tag{5}$$

$$D = \frac{\left(r_a I + \dfrac{aa^T}{1+r_a}\right)T - ab^T}{r_a^2}. \tag{6}$$

No inverse of T, D, or the full density matrix is required.

If ρ_A is pure, positivity implies $\rho = \rho_A \otimes \rho_B$. Every nonzero invertible Alice filter leaves a pure Alice marginal and the same Bob marginal. Hence

$$\boxed{B_A^*(\rho) = \|b\| \quad \text{when } \|a\| = 1,} \tag{7}$$

and the value is attained. For Bob-only filtering, interchange a, b and transpose T throughout.

Canonical data. From the canonical data (D, c) define

$$A = DD^T, \quad s = c^T c, \quad d = Dc, \quad C = [d]_\times, \quad K = A - sI, \qquad [d]_\times x = d \times x. \tag{8}$$

Thus A, K are real symmetric, while $C^T = -C$. Throughout we also use the *boundary radius*, the largest squared length of a Bloch vector to which a rank-one Alice effect steers Bob:

$$E = \max_{\|u\|=1} \left\| c + D^T u \right\|^2 ; \tag{9}$$

Section 8 evaluates it in closed form up to one scalar root.

4 The compact reduction

Idea. A filter on Alice acts on the Pauli coordinates by a Lorentz transformation followed by normalization, and the CHSH value of the output is the sum of the two largest squared singular values of its correlation matrix. Fixing the two measured directions and choosing the remaining filter freedom optimally turns the supremum over all invertible filters into the maximum of one explicit quadratic function $f(n, v)$ over a compact set of pairs of vectors in $\mathbb{R}^3$. Everything else in the paper is an analysis of this maximum.

4.1 Filtering and a Lorentz two-plane

The usual Lorentz representation of local filtering follows from the Pauli determinant identity [3]: for a Hermitian matrix $X = x_0 I + x \cdot \sigma$, $\det X = x_0^2 - \|x\|^2$. Conjugation by an invertible filter, divided by $|\det F|$, preserves this quadratic form and the future cone. Thus Alice filtering acts on the real matrix R by a Lorentz matrix, followed by trace normalization.

For a canonical marginal the denominator is the time coordinate w_0 of the timelike row. Choose the first two spatial rows to be the two maximizing directions in the Ky Fan formula for B^2; write them as (x_0, X) and (y_0, Y). Let the omitted spatial row have time coordinate z_0. Lorentz orthogonality gives

$$X^T X = 1 + x_0^2, \quad Y^T Y = 1 + y_0^2, \quad X^T Y = x_0 y_0, \quad w_0^2 = 1 + x_0^2 + y_0^2 + z_0^2. \tag{10}$$

The sum of the two selected squared correlation-row norms is

$$\frac{\mathcal{N}}{w_0^2}, \qquad \mathcal{N} = X^T A X + Y^T A Y + s(x_0^2 + y_0^2) + 2d^T(x_0 X + y_0 Y). \tag{11}$$

This numerator is a sum of squared Euclidean norms and is nonnegative.

Define

$$k = X \times Y, \quad h = x_0 Y - y_0 X, \quad \Delta = 1 + x_0^2 + y_0^2. \tag{12}$$

Elementary vector identities give

$$\|k\|^2 = \Delta, \qquad \|h\|^2 = \Delta - 1, \qquad k^T h = 0, \tag{13}$$

$$XX^T + YY^T = \Delta I - kk^T - hh^T, \qquad x_0 X + y_0 Y = -k \times h. \tag{14}$$

Set $n = k/\sqrt{\Delta}$, $v = h/\sqrt{\Delta}$. Then $\|n\| = 1$, $n^T v = 0$, and $\|v\| < 1$. Substitution gives

$$\frac{\mathcal{N}}{\Delta} = f(n, v) := \operatorname{tr} A - n^T A n - v^T K v - 2 v^T C n. \tag{15}$$

Since $w_0^2 \geq \Delta$ and $\mathcal{N} \geq 0$, every filtered CHSH value squared is at most a value of f on

$$\mathcal{D} = \{(n, v) : \|n\| = 1,\ n^T v = 0,\ \|v\| \leq 1\}. \tag{16}$$

Conversely, for any pair in $\mathcal{D}$ set $u = -n \times v$. Then $\|u\| = \|v\| \le 1$ and $n^T u = 0$. The positive filter with effect $I + u \cdot \sigma$ produces the correlation matrix

$$D_u = S_u D + u c^T, \qquad S_u = \sqrt{1 - \|u\|^2}\, I + \frac{u u^T}{1 + \sqrt{1 - \|u\|^2}}. \tag{17}$$

Because the initial marginal is maximally mixed, its normalization denominator is 1. Direct expansion, using

$$u^T A u + v^T A v = \|v\|^2 (\operatorname{tr} A - n^T A n), \tag{18}$$

shows

$$\operatorname{tr}[(I - n n^T) D_u D_u^T] = f(n, v). \tag{19}$$

The Ky Fan maximum is at least this selected-plane value. Thus

$$\boxed{(B_A^*)^2 = \max_{(n,v)\in\mathcal{D}} f(n, v).} \tag{20}$$

The boundary corresponds to rank-one filters and is reached by a limit of invertible ones. Continuity on the compact domain proves that the supremum and this maximum coincide. No Lorentz-surjectivity assertion is needed for the converse: the filter in (17) is explicit.

4.2 Boundary versus interior

On $\|v\| = 1$, the vectors $n, v, u = -n \times v$ are orthonormal, and

$$f(n, v) = s + u^T A u + 2 d^T u = \left\| c + D^T u \right\|^2. \tag{21}$$

Every unit u occurs, so the boundary maximum is exactly E.

At an interior maximum, $\|v\| < 1$. The equality-constraint gradients of $\|n\|^2 = 1$ and $n^T v = 0$ are independent: their v blocks are 0 and the nonzero vector n. Lagrange multipliers therefore exist. With a convenient sign convention their equations are

$$An - Cv = \lambda n - \beta v, \qquad Kv + Cn = -\beta n. \tag{22}$$

Take the first inner product with v and the second with n. Skew-symmetry of C and $K = A - sI$ give

$$n^T A v = -\beta \|v\|^2, \qquad n^T A v = -\beta. \tag{23}$$

Consequently $\beta(1 - \|v\|^2) = 0$, and the interior condition forces $\beta = 0$. This is the cancellation that removes the remaining continuous optimization. The stationary equations are precisely

$$\mathcal{H}\begin{pmatrix} n \\ v \end{pmatrix} = \lambda \mathcal{B}\begin{pmatrix} n \\ v \end{pmatrix}, \quad \text{or} \quad An - Cv = \lambda n, \quad Kv + Cn = 0. \tag{24}$$

Taking their inner products with n, v proves

$$f(n, v) = \operatorname{tr} A - \lambda, \qquad (\lambda - s) n^T v = 0. \tag{25}$$

5 A physical Gram matrix and the purity obstruction

The general spectral formula of Section 7 treats every value of the unilateral optimum, including sub-classical values and singular pencils. Physical positivity allows a much simpler formula throughout the entire region in which unilateral CHSH violation is possible, and we present it first. The simplification also yields polynomial activation tests and useful probability tradeoffs.

Continue to use the canonical data in (8), and write

$$\alpha = \operatorname{tr} A - s, \qquad m = \frac{\alpha - 1}{2}, \qquad H_0 = K + iC. \tag{26}$$

Here and below α denotes this purity invariant. The complex matrix H_0 is Hermitian. We use $z^\dagger z$ for the Hermitian squared norm and $z^T z$ for the complex bilinear quadratic form; the distinction will matter at the CHSH threshold.

Lemma 1 (Spin-flip Gram identity)**.** *Let*

$$\widetilde{\rho}_c = (\sigma_y \otimes \sigma_y)\rho_c^T(\sigma_y \otimes \sigma_y), \qquad S_i = \sigma_i \otimes I, \qquad G_{ij} = \operatorname{tr}(\rho_c S_i \widetilde{\rho}_c S_j).$$

Then

$$\boxed{2G = K - mI + iC \succeq 0.} \tag{27}$$

In particular $K \succeq mI$, or equivalently

$$\operatorname{tr} A + s \leq 1 + 2\lambda_{\min}(A). \tag{28}$$

For real vectors n, v and $z = n + iv$,

$$z^\dagger(K - mI + iC)z = n^T(K - mI)n + v^T(K - mI)v + 2v^T C n. \tag{29}$$

Proof. The spin flip is positive semidefinite, since it is a unitary conjugate of the transpose of a positive semidefinite matrix. For $Z = \sum_j z_j S_j$,

$$z^\dagger G z = \operatorname{tr}(\rho_c Z^\dagger \widetilde{\rho}_c Z) = \left\| \widetilde{\rho}_c^{1/2} Z \rho_c^{1/2} \right\|_F^2 \geq 0.$$

For completeness the Pauli coefficient calculation is as follows. The identity–identity contribution to G_{ij} is $\delta_{ij}/4$; the Bob-marginal contribution is $-s\delta_{ij}/4$; and the correlation–correlation contribution is $(2A_{ij} - \operatorname{tr} A\, \delta_{ij})/4$. The two remaining cross terms are

$$\frac{i}{4}\sum_k \epsilon_{ikj} d_k - \frac{i}{4}\sum_k \epsilon_{kij} d_k = -\frac{i}{2}\sum_k \epsilon_{ijk} d_k = \frac{i}{2} C_{ij}.$$

Their sum is $G = \frac{1}{2}A + \frac{1}{4}(1 - s - \operatorname{tr} A)I + \frac{i}{2}C = \frac{1}{2}(K - mI + iC)$. Expanding $z = n + iv$ and using $C^T = -C$ gives (29). Taking $v = 0$ gives the real-matrix inequality $K \succeq mI$. □

The bare implication from CHSH violation to a negative conditional Rényi-2 entropy is established in the literature [10]. The next proof records its precise quantitative form and its covariance under unilateral filtering, because these are the ingredients needed for the spectral simplification. The plane-purity identity (33) and the determinant covariance (30), with the roles of the two parties exchanged, also appear in the proof of Conjecture 1 of Milne, Jennings, Jevtic and Rudolph recorded in the Omega Institute's public repository [16] (Section 2); they are not claimed as new here.

Proposition 2 (Purity obstruction and determinant covariance)**.** *For a possibly unnormalized Hermitian two-qubit matrix X, define*

$$\Delta_B(X) = \operatorname{tr} X^2 - \operatorname{tr}[(\operatorname{tr}_A X)^2].$$

For every complex 2×2 matrix F,

$$\Delta_B((F \otimes I)X(F^\dagger \otimes I)) = |\det F|^2 \Delta_B(X). \tag{30}$$

For every normalized two-qubit state,

$$B(\rho)^2 \le 1 + 2\Delta_B(\rho). \tag{31}$$

Consequently Alice-only CHSH violation requires $\Delta_B(\rho) > 0$. For a canonical state and a filter with effect proportional to $I + u \cdot \sigma$,

$$B((\rho_c)_u)^2 \le 1 + m(1 - \|u\|^2). \tag{32}$$

Thus $B_A^* > 1$ implies $\alpha > 1$ and $K \succ 0$.

Proof. Expand $X = \frac{1}{2}\sum_{\nu=0}^{3} X_\nu \otimes \sigma_\nu$, with Hermitian 2×2 coefficients X_ν. Pauli orthogonality and $\operatorname{tr} Y^2 - (\operatorname{tr} Y)^2 = -2 \det Y$ give

$$\Delta_B(X) = \frac{1}{2}\sum_{\nu}[\operatorname{tr}(X_\nu^2) - (\operatorname{tr} X_\nu)^2] = -\sum_{\nu} \det X_\nu.$$

Every coefficient transforms as $X_\nu \mapsto F X_\nu F^\dagger$, which proves (30), including singular F.

For the quantitative bound, let (a, b, T) be the Bloch data of ρ, let n be a real unit vector, and conjugate the full spin flip $\widetilde{\rho}$ by $(n \cdot \sigma) \otimes I$ to obtain the positive matrix $\widetilde{\rho}_n$. Conjugation on Alice acts on Bloch vectors by $2nn^T - I$, so Pauli orthogonality gives

$$4 \operatorname{tr}(\rho\widetilde{\rho}_n) = 1 + \|a\|^2 - \|b\|^2 - \operatorname{tr}(TT^T) - 2(a \cdot n)^2 + 2n^T TT^T n.$$

Together with $4\Delta_B(\rho) = \|a\|^2 - \|b\|^2 + \operatorname{tr}(TT^T) - 1$, this proves the exact identity

$$\operatorname{tr}(TT^T) - n^T TT^T n + (a \cdot n)^2 + 2\operatorname{tr}(\rho\widetilde{\rho}_n) = 1 + 2\Delta_B(\rho). \tag{33}$$

Both of the last two terms on the left are nonnegative. Minimizing $n^T TT^T n$ proves (31). Equality requires a maximizing CHSH plane whose omitted direction has $a \cdot n = 0$ and $\operatorname{tr}(\rho\widetilde{\rho}_n) = 0$.

If a filter succeeds with probability p, covariance and normalization give

$$B(\rho_F)^2 \le 1 + \frac{2|\det F|^2}{p^2}\Delta_B(\rho).$$

In canonical coordinates, $\Delta_B(\rho_c) = (\alpha - 1)/4$, and the effect $I + u \cdot \sigma$ has normalization one and determinant $1 - \|u\|^2$. This proves (32). The same inequality with $\alpha \le 1$ excludes every violation, including any limiting violation. If $\alpha > 1$, Lemma 1 gives $K \succeq mI \succ 0$. □

For the original Bloch data, put $h = 1 - \|a\|^2 > 0$. Whitening has $|\det W|^2 = 1/h$, so covariance gives the useful identities

$$\boxed{\alpha - 1 = \frac{\|T\|_F^2 + \|a\|^2 - \|b\|^2 - 1}{h}, \qquad \alpha = \frac{\|T\|_F^2 - \|b\|^2}{h}.} \tag{34}$$

Here $\|T\|_F$ is the Frobenius norm. Thus the purity guard is already a quadratic polynomial inequality in the original state coordinates.

6 The smallest eigenvalue is automatically feasible

Theorem 3 (A single smallest eigenvalue in the violation-relevant region)**.** *For a physical canonical state with $\alpha > 1$, define $M = A + CK^{-1}C$. Then*

$$\boxed{(B_A^*)^2 = \operatorname{tr} A - \lambda_{\min}(M).} \tag{35}$$

Every unit vector n in the minimum eigenspace gives an optimal invertible filter:

$$v = -K^{-1}Cn, \qquad u = -n \times v, \qquad F \text{ proportional to } \sqrt{I + u \cdot \sigma}\, W. \tag{36}$$

In particular, $n \cdot v = 0$ and $\|v\| < 1$ hold automatically. All optimal pairs in the compact reduction are precisely the pairs in (36).

The same conclusion holds under the algebraic hypothesis $K + iC \succ 0$, even when $\alpha \le 1$.

Proof. The Gram identity gives $H_0 \succeq mI \succ 0$ when $\alpha > 1$. Under either hypothesis in the theorem, the real representation

$$\begin{pmatrix} K & -C \\ C & K \end{pmatrix}$$

is positive definite. Its Schur complement is $L = K + CK^{-1}C = M - sI \succ 0$. Let $\delta = \lambda_{\min}(L) > 0$, choose $\|n\| = 1$ with $Ln = \delta n$, and put $v = -K^{-1}Cn$. Then

$$Kn - Cv = \delta n, \qquad Kv + Cn = 0. \tag{37}$$

Taking the inner product of the first equation with v and of the second with n gives $\delta n \cdot v = 0$. Hence $n \cdot v = 0$.

The crucial norm estimate is not a generic property of a stationary eigenvector. It follows here from the following exact identity:

$$\boxed{v^T L v = \delta - \delta^2 n^T K^{-1} n.} \tag{38}$$

Indeed $Cv = (K - \delta I)n$, and taking the inner product of the first equation in (37) with n yields $v^T K v = n^T K n - \delta$. Therefore

$$\begin{aligned} v^T L v &= v^T K v - (Cv)^T K^{-1}(Cv) \\ &= n^T K n - \delta - n^T (K - \delta I) K^{-1} (K - \delta I) n \\ &= \delta - \delta^2 n^T K^{-1} n. \end{aligned}$$

The minimum Rayleigh bound for L now gives

$$\|v\|^2 \leq 1 - \delta n^T K^{-1} n < 1. \tag{39}$$

Thus the constructed pair belongs to the interior of the exact compact domain (16).

For the matching upper bound, complete the square in (15):

$$f(n, v) = \operatorname{tr} A - n^T M n - (v + K^{-1}Cn)^T K (v + K^{-1}Cn). \tag{40}$$

The last term is nonnegative, and $n^T M n \geq \lambda_{\min}(M)$. The constructed pair gives equality in both inequalities. Because $K \succ 0$, equality also characterizes every optimizing pair: n must lie in the minimum eigenspace and $v = -K^{-1}Cn$. □

Corollary 4 (Strict boundary gap and a uniform probability guarantee)**.** *Whenever $H_0 \succ 0$, the optimum strictly exceeds the rank-one boundary value: $(B_A^*)^2 > E$, even if the optimum is at most one. Writing $\delta = \lambda_{\min}(M) - s > 0$, every canonical optimal effect obeys*

$$\|u\|^2 \leq 1 - \frac{\delta}{\lambda_{\max}(K)}, \qquad p_{\max} \geq \frac{1}{1 + \sqrt{1 - \delta/\lambda_{\max}(K)}} > \frac{1}{2}. \tag{41}$$

If the minimum eigenvalue of M is simple, the positive optimal filter is unique up to scale. An arbitrary output unitary remains irrelevant to the CHSH value.

Proof. Every equality pair in (40) satisfies the strict estimate (39). No boundary pair can therefore attain the global maximum. Compactness of the boundary gives the strict gap. Moreover $n^T K^{-1} n \geq 1/\lambda_{\max}(K)$ gives the first inequality in (41); its radicand is nonnegative because $M - sI = K + CK^{-1}C \preceq K$. The probability statement follows by maximal Kraus rescaling. For a simple eigenvalue, n is unique up to sign, and simultaneous sign change of n, v leaves $u = -n \times v$ fixed. The equality discussion following (65) shows that this reconstruction recovers the original positive-filter part of every finite optimum. □

Corollary 5 (Sharp purity ceiling and its equality cases)**.** *If $\alpha > 1$, then*

$$(B_A^*)^2 \leq \frac{1+\alpha}{2}. \tag{42}$$

Equality holds exactly when

$$\ker(K - mI) \cap \ker C \neq \{0\}. \tag{43}$$

In that case the canonical state itself is optimal. Every state of rank at least three satisfies (42) strictly.

Proof. Apply (32) to the optimal filter supplied by Theorem 3. Since $m > 0$, equality forces its canonical effect to have $u = 0$. The omitted direction n then satisfies $n^T(K - mI)n = 0$. Positivity of $K - mI + iC$ implies that this matrix annihilates n; its real and imaginary parts yield $(K - mI)n = 0$ and $Cn = 0$. Conversely, such a unit vector gives $f(n, 0) = \alpha - m = (1 + \alpha)/2$.

For the rank statement, a real nonzero n gives an invertible $Z = (n \cdot \sigma) \otimes I$. If $r = \operatorname{rank} \rho_c \geq 3$, then

$$\operatorname{rank}(\widetilde{\rho}_c^{1/2} Z \rho_c^{1/2}) \geq 2r - 4 > 0.$$

Thus its squared Frobenius norm cannot vanish, excluding a real nonzero vector in $\ker G$. More generally, the same argument excludes any $z \in \ker G$ with $z^T z \neq 0$, since $\det(z \cdot \sigma) = -z^T z$. □

6.1 Why positivity of K alone is insufficient

The purity or positive-Gram hypothesis in Theorem 3 cannot be replaced by $K \succ 0$. Consider the full-rank canonical state

$$\rho_{\text{guard}} = \frac{1}{40} \begin{pmatrix} 19 & 0 & 0 & 7 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 7 & 0 \\ 7 & 0 & 0 & 13 \end{pmatrix}. \tag{44}$$

Its parameters are

$$\begin{aligned} c &= (0, 0, 3/10), & D &= \operatorname{diag}(7/20, -7/20, 3/5), \\ K &= \operatorname{diag}(13/400, 13/400, 27/100) \succ 0, & \alpha &= 103/200. \end{aligned}$$

The unrestricted smallest-eigenvalue expression gives

$$\operatorname{tr} A - \lambda_{\min}(M) = \frac{7693}{5200} > 1, \qquad \|v\| = \frac{72}{13} > 1.$$

The spurious minimum eigenspace is the entire transverse plane, and every unit vector in that plane has the same infeasible norm. The only remaining interior eigenvalue gives $49/200$. For the boundary radius, writing $z = u_3$ yields a convex quadratic on $[-1, 1]$ whose largest endpoint is $(3/10 + 3/5)^2 = 81/100$. The general spectral theorem therefore gives the exact answer

$$\boxed{(B_A^*(\rho_{\text{guard}}))^2 = 81/100.}$$

This example explains why an unguarded cubic formula can incorrectly predict a CHSH violation.

6.2 A full-rank violating state with a circle of optimal filters

The simple-eigenvalue hypothesis in the uniqueness statement cannot be discarded, even for a full-rank state with a strictly violating optimum. Take

$$c = (9/20, 0, 0), \qquad D = \operatorname{diag}(8/15, -2/3, 3/4),$$

or equivalently

$$\rho_{\text{circle}} = \begin{pmatrix} 7/16 & 9/80 & 0 & 3/10 \\ 9/80 & 1/16 & -1/30 & 0 \\ 0 & -1/30 & 1/16 & 9/80 \\ 3/10 & 0 & 9/80 & 7/16 \end{pmatrix}. \tag{45}$$

In the Bell-parity basis its two blocks are $\left(\begin{smallmatrix} 59/80 & 9/80 \\ 9/80 & 7/240 \end{smallmatrix}\right)$ and $\left(\begin{smallmatrix} 11/80 & 9/80 \\ 9/80 & 23/240 \end{smallmatrix}\right)$. Their determinants are $17/1920$ and $1/1920$, respectively, and their diagonals are positive, giving an exact certificate that $\rho_{\text{circle}} \succ 0$. Here

$$\alpha = \frac{49}{45}, \qquad d = (6/25, 0, 0),$$
$$K = \operatorname{diag}\left(\frac{59}{720}, \frac{871}{3600}, \frac{9}{25}\right),$$
$$M - sI = \operatorname{diag}\left(\frac{59}{720}, \frac{59}{720}, \frac{531}{4355}\right).$$

The minimum eigenspace is $\operatorname{span}(e_1, e_2)$. For $n = (\cos\theta, \sin\theta, 0)$, $v = (0, 0, -2\sin\theta/3)$, so the complete set of optimal canonical effects is

$$u = \frac{2}{3}(\sin^2\theta, -\cos\theta\sin\theta, 0), \qquad (u_1 - \tfrac{1}{3})^2 + u_2^2 = \frac{1}{9}, \quad u_3 = 0. \tag{46}$$

Every point of this circle attains $(B_A^*)^2 = 145/144 > 1$. The circle includes the identity effect $u = 0$, and all its points have $\|u\| \le 2/3 < 1$. Thus equality cases can form a genuine continuous family of distinct positive filters without loss of rank or CHSH violation.

7 All states: finite spectral candidates

Idea. Outside the guard region of Section 6 the maximum of f may sit on the boundary of the compact domain (a rank-one limit of invertible filters) or at one of finitely many interior stationary points. The interior candidates are the roots of a cubic pencil determinant, and the boundary candidate is the radius E. This section states the resulting formula for every state, including all singular and degenerate cases; its proof is given in Appendix A.

For the canonical data (8), introduce three real symmetric 6×6 matrices

$$\mathcal{H} = \begin{pmatrix} A & -C \\ C & K \end{pmatrix}, \qquad \mathcal{B} = \begin{pmatrix} I & 0 \\ 0 & 0 \end{pmatrix}, \qquad \mathcal{J} = \begin{pmatrix} I & 0 \\ 0 & -I \end{pmatrix}. \tag{47}$$

Let

$$\mathcal{L}_0 = \ker\mathcal{H} \cap \ker\mathcal{B} = \{(0, w) : w \in \ker K \cap \ker C\}. \tag{48}$$

Choose any orthonormal-column matrix U spanning $\mathcal{L}_0^{\perp}$, and put

$$\widehat{H} = U^T\mathcal{H}U, \qquad \widehat{B} = U^T\mathcal{B}U, \qquad p(z) = \det(\widehat{H} - z\widehat{B}). \tag{49}$$

The choice of U does not affect the following construction.

For each distinct real root λ of p, choose an orthonormal-column basis Z_λ of $\ker(\widehat{H} - \lambda\widehat{B})$, set $Y_\lambda = UZ_\lambda$, and define

$$\chi_\lambda = \lambda_{\max}(Y_\lambda^T\mathcal{J}Y_\lambda). \tag{50}$$

These are finite linear-algebra operations. A level is *admissible* when $\chi_\lambda \ge 0$.

Finally let E be the boundary radius (9), given explicitly in Section 8.

Theorem 6 (Unilateral spectral formula)**.** *The polynomial p is nonzero, has degree at most three, and every one of its roots is real. Each eigenspace used in (50) has dimension at most three. For every two-qubit state with invertible Alice marginal,*

$$\boxed{\left(B_A^*(\rho)\right)^2 = \max\left(\{E\} \cup \{\operatorname{tr} A - \lambda : p(\lambda) = 0,\ \chi_\lambda \geq 0\}\right).} \tag{51}$$

This remains valid when ρ, D, A, K, or the pencil blocks are singular, and when eigenvalues are repeated. Equation (7) covers the only excluded marginal case. The Bob formula follows by interchanging parties.

The result requires at most three interior candidate values, not an optimization over filter directions or measurement settings. If the data are algebraic, all steps are exact algebraic operations with decidable comparisons. The word "closed" here means this finite algebraic/spectral expression, not a universal expression in radicals for the boundary sextic.

7.1 Generic form: one symmetric 3×3 matrix

When K is invertible, no common kernel is present and the Schur complement gives

$$p(z) = \det K \det(M - zI), \qquad \boxed{M = A + CK^{-1}C.} \tag{52}$$

For each eigenspace $\mathcal{E}_\lambda = \ker(M - \lambda I)$, choose an orthonormal basis matrix N_λ. Then admissibility is exactly

$$\boxed{\sigma_{\min}(K^{-1}CN_\lambda) \leq 1.} \tag{53}$$

For a simple eigenvalue with unit eigenvector n, this is just $\|K^{-1}Cn\| \leq 1$. Use strict inequality for an invertible-filter representative at that level. A repeated eigenspace must be tested as a whole; testing an arbitrary eigenbasis vector by vector can give the wrong answer.

An explicit cubic is also available. Put $t = z - s$, $\delta = \det K$, $q = d^T K d$, $r = d^T d$, and $e_2 = ((\operatorname{tr} K)^2 - \operatorname{tr} K^2)/2$. Before common-kernel removal,

$$\begin{aligned} \det(\mathcal{H} - z\mathcal{B}) = & -\delta t^3 + [\delta \operatorname{tr} K + d^T \operatorname{adj}(K) d - e_2 r] t^2 \\ & - (\delta - q)(e_2 - r)t + (\delta - q)^2. \end{aligned} \tag{54}$$

This polynomial identity holds also for singular K. If it vanishes identically because of a common kernel, use (49); do not attempt to select roots of the zero polynomial.

For $\lambda \neq s$, a feasible unit vector n gives

$$v = -K^{-1}Cn, \qquad u = -n \times v. \tag{55}$$

Orthogonality $n^T v = 0$ is automatic. At $\lambda = s$, apply the phase operation in Section A.2 before forming u.

7.2 An ordinary 3×3 compression for singular K

The pencil is conceptually convenient, but an implementation can use only ordinary symmetric eigenproblems of dimension at most three. Let N have orthonormal columns spanning $\ker K$, set $L = CN$, and let P have orthonormal columns spanning $(\operatorname{ran} L)^\perp$. Define

$$M_0 = A + CK^\dagger C, \qquad M_L = P^T M_0 P. \tag{56}$$

For an eigenvalue λ of M_L, let Q be an ambient orthonormal basis of its eigenspace in $\operatorname{ran} P$. Put

$$V_\lambda = -K^\dagger C Q + N L^\dagger (M_0 - \lambda I) Q. \tag{57}$$

For $\lambda \neq s$, the branch is admissible exactly when $\lambda_{\min}(V_\lambda^T V_\lambda) \leq 1$. Strict inequality gives an interior representative. For $\lambda = s$, use the complex-kernel prescription below; the closed branch is always feasible. The reference implementation uses this compression.

8 The boundary radius, including the hard case

Diagonalize A: let its eigenvalues be $\alpha_i \geq 0$, and write the coordinates of d in an orthonormal eigenbasis as g_i. Set $\alpha_* = \max_i \alpha_i$.

If the component of d in the top eigenspace is zero and

$$h = \sum_{\alpha_i < \alpha_*} \frac{g_i^2}{(\alpha_* - \alpha_i)^2} \leq 1, \tag{58}$$

then the hard-case answer is

$$\boxed{E = s + \alpha_* + \sum_{\alpha_i < \alpha_*} \frac{g_i^2}{\alpha_* - \alpha_i}.} \tag{59}$$

An extremizing u has coordinates $u_i = g_i/(\alpha_* - \alpha_i)$ below the top eigenspace, and an arbitrary top component of squared norm $1 - h$.

In every other case, there is a unique $\eta > \alpha_*$ satisfying

$$\sum_i \frac{g_i^2}{(\eta - \alpha_i)^2} = 1. \tag{60}$$

Then

$$\boxed{E = s + \eta + \sum_i \frac{g_i^2}{\eta - \alpha_i}, \qquad u = (\eta I - A)^{-1} d.} \tag{61}$$

This is a specified root of a polynomial of degree at most six:

$$\det(\eta I - A)^2 - d^T \operatorname{adj}(\eta I - A)^2 d = 0, \qquad \eta > \alpha_*. \tag{62}$$

Use the hard-case rule rather than accepting poles or arbitrary roots of this polynomial.

Proof of the boundary formula. The squared radius is $s + u^T A u + 2 d^T u$ on the unit sphere. In both constructions, $H = \eta I - A \succeq 0$, $Hu = d$, and $\|u\| = 1$. For any other unit vector x,

$$(s + u^T A u + 2 d^T u) - (s + x^T A x + 2 d^T x) = (x - u)^T H (x - u) \geq 0. \tag{63}$$

Thus the constructed value is globally optimal. Equality means $x - u \in \ker H$ together with $\|x\| = 1$. In the regular case, the secular function is strictly decreasing from a value greater than 1 (or infinity) to 0, proving existence and uniqueness. In the hard case, the specified top component supplies exactly the missing norm. The formulas also cover $d = 0$, repeated eigenvalues, $A = 0$, and a point ellipsoid. □

For physical canonical states $E \leq 1$: conditioning Bob on any rank-one Alice effect produces a valid qubit Bloch vector $c + D^T u$.

9 Optimal filters, attainment and violation criteria

Let S denote the right side of (51). For an admissible level $\operatorname{tr} A - \lambda = S$, choose a normalized feasible solution (n, v) as above and put $u = -n \times v$. An extremizing filter on the original state is

$$\boxed{F = \kappa V \sqrt{I + u \cdot \sigma}\, (2\rho_A)^{-1/2},} \tag{64}$$

where V is any output unitary and $\kappa > 0$ is small enough that $F^\dagger F \leq I$. Choosing $\kappa = 1/\left\| \sqrt{I + u \cdot \sigma} (2\rho_A)^{-1/2} \right\|_{\mathrm{op}}$ maximizes the success probability for this normalized branch. Its success probability is κ^2, since the unscaled combined filter has trace normalization 1.

If $\|v\| < 1$, the filter is invertible and attains the value. If $\|v\| = 1$, it is rank one and yields a limiting value for the paper's invertible-filter domain. Likewise the boundary maximizers from Section 8 give rank-one filters, approached by replacing u with ru, $r \uparrow 1$.

Corollary 7 (Exact attainment criterion)**.** *For an invertible filtered-side marginal and $S > 0$, the supremum is attained by an invertible filter if and only if there is a root λ of p such that*

$$\operatorname{tr} A - \lambda = S, \qquad \chi_\lambda > 0. \tag{65}$$

If $S > E$, attainment is automatic. If $S = 0$, necessarily $D = c = 0$, and every invertible filter attains zero. The pure-marginal case is given by (7).

Proof. Sufficiency follows from the explicit construction. Conversely, a positive finite optimum must have equality in the Lorentz denominator bound. The numerator is then positive, so $z_0 = 0$ in (10). It therefore supplies an interior maximizing pair in $\mathcal{D}$, which satisfies (24) and has positive $\mathcal{J}$-quadratic form. Lorentz completeness gives $w = (e_0 + x_0(x_0, X) + y_0(y_0, Y))/\sqrt{\Delta}$, hence its effect direction is $w_{\mathrm{sp}}/w_0 = (x_0 X + y_0 Y)/\Delta = -n \times v$. Thus reconstruction recovers the original positive-filter part, not merely some other optimum. If $S > E$, a compact maximizer cannot be on the boundary. Finally $E = 0$ implies $c + D^T u = 0$ for every unit u; comparing u and $-u$ gives $c = 0$ and $D = 0$. □

For positive S, this also characterizes *all* invertible optimal filters up to irrelevant scale and output unitary: take all stationary solutions at the maximizing level with $\|n\| = 1$, $n^T v = 0$, $\|v\| < 1$, and use (64). All rank-one optimal filters are the boundary-radius solutions when $S = E$.

Corollary 8 (Necessary and sufficient unilateral violation criterion)**.** *For a physical state with invertible filtered-side marginal,*

$$\boxed{B_A^*(\rho) > 1 \quad \Longleftrightarrow \quad \exists \lambda : \; p(\lambda) = 0, \quad \lambda < \operatorname{tr} A - 1, \quad \chi_\lambda > 0.} \tag{66}$$

Thus Alice-only activation from a CHSH-nonviolating state occurs exactly when (66) holds and $s_1(T)^2 + s_2(T)^2 \leq 1$ for the original state. The Bob criterion is obtained by swapping parties. No ellipsoid-centre conjecture is used.

Proof. The boundary obeys $E \leq 1$, so any value above 1 must be an attained interior spectral value. Conversely a feasible level exceeding 1 supplies a violating filter. A level exceeding 1 cannot have only a boundary representative, which explains the strict eligibility condition. □

Corollary 9 (One-party maximal violation)**.** $B_A^*(\rho) = \sqrt{2}$ *if and only if ρ is pure and entangled. The same is true for Bob.*

Proof. A pure entangled state can be converted to a Bell state by whitening either marginal. Conversely a value $\sqrt{2} > 1$ must be attained by an invertible filter. Every singular value of a physical correlation matrix is at most 1, so $B^2 = 2$ forces two singular values to equal 1. Their singular vectors give two orthogonal perfect correlation directions. The corresponding commuting Pauli-product stabilizers have a joint eigenspace projector of trace one, hence a one-dimensional simultaneous eigenspace, a Bell state up to local unitaries. Thus the filtered state is pure maximally entangled; invertible local filtering preserves density rank and Schmidt rank, so the original state is pure and entangled. □

10 Root-free tests in the original state coordinates

Theorem 10 (Polynomial and ellipsoid activation criteria)**.** *For a state with mixed Alice marginal, Alice-only CHSH violation is equivalent to*

$$\alpha > 1 \quad \textit{and} \quad M - (\operatorname{tr} A - 1)I \not\succeq 0. \tag{67}$$

Let $t = \operatorname{tr} A - 1$*. Within the guarded region* $\alpha > 1$*, define*

$$R_A = \det K\,(A - tI) + C\operatorname{adj}(K)C. \tag{68}$$

Then the second condition is equivalent to $R_A \not\succeq 0$*.*

An orientation-independent expression uses only the steering ellipsoid shape $Q = D^T D$ *and center* c*. Put*

$$s = c^T c, \qquad \delta_K = \det(Q - sI), \qquad q = c^T Q(Q - sI)c, \qquad t = \operatorname{tr} Q - 1,$$

and define the real symmetric polynomial matrix

$$\boxed{\begin{aligned} R_E = {} & \det Q\,(\delta_K - q)I + (sq - t\delta_K)\operatorname{adj} Q \\ & + \det Q\,(Q - sI)cc^T(Q - sI). \end{aligned}} \tag{69}$$

Activation is equivalent to $\operatorname{tr} Q - s > 1$ *and* $R_E \not\succeq 0$*. For either* $R = R_A$ *or* $R = R_E$*, failure of positive semidefiniteness is equivalent to at least one of the following three numbers being negative:*

$$e_1(R) = \operatorname{tr} R, \qquad e_2(R) = \frac{(\operatorname{tr} R)^2 - \operatorname{tr} R^2}{2}, \qquad e_3(R) = \det R. \tag{70}$$

Proof. The purity obstruction excludes $\alpha \le 1$; for $\alpha > 1$, Theorem 3 gives (67). Here $\det K > 0$, and $R_A = \det K\,[M - tI]$, proving the first polynomial test.

The cross-product identity

$$CK^{-1}C = \frac{Kdd^TK - (d^TKd)K}{\det K} \tag{71}$$

can be checked after diagonalizing the positive matrix K; for example its $(1,1)$ entry is $-d_3^2/k_2 - d_2^2/k_3$, and its $(1,2)$ entry is $d_1 d_2/k_3$. Under the guard, Q and $Q - sI$ are positive definite, so D is invertible, $\det K = \delta_K > 0$, and $d^TKd = q$. Substitution gives

$$\delta_K(M - tI) = (\delta_K - q)A + (sq - t\delta_K)I + Kdd^TK.$$

Congruence by D^{-1} and D^{-T} therefore gives

$$D^{-1}\delta_K(M - tI)D^{-T} = (\delta_K - q)I + (sq - t\delta_K)Q^{-1} + (Q - sI)cc^T(Q - sI). \tag{72}$$

Multiplication by the positive number $\det Q$ produces R_E. Congruence and positive scaling preserve positive semidefiniteness.

Finally, let $\lambda_1, \lambda_2, \lambda_3$ be the real eigenvalues of a symmetric matrix R. If all are nonnegative, their elementary symmetric coefficients are nonnegative. Conversely, if all three coefficients in (70) are nonnegative, then $x^3 + e_1x^2 + e_2x + e_3$ is strictly positive for every $x > 0$. Since this polynomial is $\prod_j(x + \lambda_j)$, it cannot have the positive root associated with a negative λ_j. □

All data in the ellipsoid test can be obtained rationally from the original Bloch coefficients. With $h = 1 - \|a\|^2 > 0$ and $X = T - ab^T$,

$$c = \frac{b - T^Ta}{h}, \qquad Q = \frac{1}{h}X^T\left(I + \frac{aa^T}{h}\right)X. \tag{73}$$

For example, the matrix multiplying X in the formula for D is $L = \sqrt{h}\, I + aa^T/(1+\sqrt{h})$, and $L^2 = hI + aa^T$. This proves (73) and removes its apparent square roots. Thus the activation decision consists of rational arithmetic and three sign comparisons, after the explicit quadratic guard. Clearing positive powers of h makes it a polynomial decision on the original state data.

Corollary 11 (A rational cubic for the optimum itself)**.** *In the region* $\operatorname{tr} Q - s > 1$*, retain* δ_K, q *from Theorem 10, and put*

$$z = (Q - sI)c, \qquad J = (\delta_K - q)(Q - sI).$$

The cubic

$$P(\nu) = \det(\nu I - J) - z^T Q \operatorname{adj}(\nu I - J) z \tag{74}$$

has three real positive roots, counted with multiplicity. If $\nu_{\min}$ *is the smallest, then*

$$\boxed{(B_A^*)^2 = \alpha - \frac{\nu_{\min}}{\delta_K}.} \tag{75}$$

The coefficients are rational functions of the original Bloch data, so this value formula requires no choice of a square root of Q*.*

Proof. Since $Kd = Dz$, the cross-product identity gives the similarity

$$D^{-1}\delta_K(M - sI)D = J + zz^T Q.$$

The right side is therefore similar to a positive definite symmetric matrix and has three real positive eigenvalues. The rank-one determinant identity gives $\det(\nu I - J - zz^T Q) = \det(\nu I - J) - z^T Q \operatorname{adj}(\nu I - J) z$, including values where $\nu I - J$ is singular. Thus its least eigenvalue is $\delta_K[\lambda_{\min}(M) - s]$, and (75) follows from the shortcut theorem. □

Corollary 12 (Directional guards and hidden activation)**.** *Define*

$$G_A = \|T\|_F^2 + \|a\|^2 - \|b\|^2 - 1, \qquad G_B = \|T\|_F^2 + \|b\|^2 - \|a\|^2 - 1.$$

Alice activation requires $G_A > 0$*; Bob activation requires* $G_B > 0$*. If* $\|T\|_F^2 \le 1$*, at most one party can activate, and it must be the party with the longer local Bloch vector. Both parties activating requires* $\|T\|_F^2 > 1$*.*

The additional condition that the original state does not violate CHSH is exactly

$$TT^T - (\operatorname{tr}(TT^T) - 1)I \succeq 0. \tag{76}$$

Combining its three nonnegative coefficients from (70), the appropriate guard, and (69) gives a necessary-and-sufficient algebraic test for hidden *unilateral activation.*

Proof. The guards follow from (34) and the swapped formula. Their sum is $2(\|T\|_F^2 - 1)$; when this is nonpositive, both cannot be positive. If $\|a\| \le \|b\|$ and $\|T\|_F^2 \le 1$, then $G_A \le 0$, proving the directional assertion. Finally $B(\rho)^2 = \operatorname{tr}(TT^T) - \lambda_{\min}(TT^T)$, which is at most one precisely under (76). □

11 Exact equality at the local CHSH bound

The strict activation test has a complete companion classification at equality. It is particularly useful when the generic matrix K is singular and the optimum is a rank-one limit.

Theorem 13 (Three regimes at $B_A^* = 1$)**.** *Assume the Alice marginal is mixed and put* $S = (B_A^*)^2$*.*

(i) If $\alpha < 1$, then $S \le 1$, and $S = 1$ if and only if $E = 1$. Equality is never attained by an invertible filter.

(ii) If $\alpha = 1$, then

$$S = 1 \iff \det H_0 = 0 \iff \det K = d^T K d.$$

Equality is attained by an invertible filter if and only if $\ker H_0$ contains z with $z^T z \neq 0$.

(iii) If $\alpha > 1$, the optimum is attained by an invertible filter and

$$S < 1 \iff R_A \succ 0, \qquad S = 1 \iff R_A \succeq 0,\ \det R_A = 0, \qquad S > 1 \iff R_A \not\succeq 0.$$

The same statements hold for Bob after swapping the parties.

Proof. When $m < 0$, the Gram identity and the compact reduction give $f(n, v) \le 1 + m(1 - \|v\|^2)$. Every interior pair has value strictly less than one. Hence a compact maximizer of value one must lie on the boundary, where the maximum is E. Equivalently, the actual-filter purity bound excludes finite attainment because every invertible filter has positive determinant.

When $m = 0$, $H_0 = 2G \succeq 0$ and, using $\|n\| = 1$,

$$f(n, v) = 1 - (n + iv)^\dagger H_0 (n + iv). \tag{77}$$

If H_0 is nonsingular, its smallest eigenvalue is positive and the subtracted term is uniformly positive on the compact domain. If $0 \neq z \in \ker H_0$, multiply z by a phase so that $\Re z \cdot \Im z = 0$ and $\|\Re z\| \ge \|\Im z\|$. The real part is then nonzero. Divide by its norm to get a feasible pair (n, v) with value one. The norm is strict exactly when the original kernel contains a vector with $z^T z \neq 0$. The identity $\det(K + iC) = \det K - d^T K d$ is polynomial in K, d: it follows from diagonal K and then orthogonal invariance, so it also holds for singular K.

The final regime follows directly from Theorem 3, since $\det K > 0$ and $R_A = \det K[M - (\operatorname{tr} A - 1)I]$. □

Proposition 14 (Boundary equality, product kernels, and rank)**.** *For a two-qubit state with mixed filtered-side marginal,*

$$E = 1 \iff \ker \rho \text{ contains a nonzero product vector.} \tag{78}$$

In particular, every state of rank at most two has $E = 1$. A rank-three state has $E = 1$ precisely when its one-dimensional kernel is generated by a product vector. At $\alpha = 1$, a state of rank at least three cannot attain $S = 1$ by an invertible filter.

Proof. A rank-one Alice effect onto $|x\rangle$ produces the conditional Bob matrix $\langle x|\rho|x\rangle$. It has positive trace because ρ_A is invertible. This 2×2 matrix is pure after normalization exactly when it has a nonzero vector $|y\rangle$ in its kernel. Positivity makes $\langle x, y|\rho|x, y\rangle = 0$ equivalent to $|x\rangle \otimes |y\rangle \in \ker \rho$. The statement is unchanged by invertible whitening, which maps product vectors to product vectors.

If $\operatorname{rank} \rho \le 2$, the kernel contains a two-dimensional complex subspace. Identify two independent vectors of this subspace with 2×2 coefficient matrices U, V. The homogeneous quadratic $\det(sU + tV)$ either vanishes identically or has a root in the complex projective line. The associated nonzero matrix has rank one, and hence represents a product vector. The rank-three assertion is immediate.

At $\alpha = 1$, finite equality requires a nonisotropic vector in $\ker H_0 = \ker G$. The rank argument at the end of Corollary 5 excludes such a vector when the rank is at least three. □

Theorem 15 (Complete rational formula for rank at most two)**.** *If* $\operatorname{rank}\rho \le 2$ *and the filtered-side marginal is mixed, then*

$$\boxed{(B_A^*)^2 = \max\left\{1, \frac{1+\alpha}{2}\right\}.} \tag{79}$$

In particular, the purity guard is both necessary and sufficient for unilateral violation in this rank range. The optimum is attained by an invertible filter exactly when $\alpha \ge 1$*, and whitening alone attains it throughout that region. For* $\alpha > 1$*, whitening is the unique positive optimal filter up to scale, with maximal success probability* $1 - \|a\|$*.*

In the original state coordinates, let $J_\rho = 1 + \|T\|_F^2 - \|a\|^2 - \|b\|^2$ *and* $G_A = \|T\|_F^2 + \|a\|^2 - \|b\|^2 - 1$*. Then*

$$\boxed{(B_A^*)^2 = \max\left\{1, \frac{J_\rho}{2(1-\|a\|^2)}\right\} = 1 + \frac{\max(0, G_A)}{2(1-\|a\|^2)}.} \tag{80}$$

When both marginals are mixed, the larger local Bloch-vector length gives a weakly larger unilateral optimum.

Proof. The canonical state is the normalized Choi matrix of a trace-preserving qubit channel. Theorem 12 and equation (17) of Ruskai–Szarek–Werner [11] show that every such channel of Choi rank at most two is unitarily equivalent to one with translation $(0, 0, \sin u \sin v)$ and diagonal linear part $(\cos u, \cos v, \cos u \cos v)$. This theorem includes the unital, non-extreme rank-two channels. Accounting for the sign of σ_y^T in the Choi correspondence, and using local rotations to choose signs, our state therefore has the form

$$D = \operatorname{diag}(\sqrt{x}, -\sqrt{y}, \sqrt{xy}), \qquad c = (0, 0, \sqrt{(1-x)(1-y)}), \qquad 0 \le x, y \le 1. \tag{81}$$

Thus $\alpha = 2(x+y) - 1$, and the canonical CHSH value is $B(\rho_c)^2 = x + y$, because $xy \le x, y$.

If $\alpha > 1$, the purity ceiling is $(1+\alpha)/2 = x + y$ and is already achieved by the canonical state. If $\alpha = 1$, the purity bound is one and the canonical state again achieves it. If $\alpha < 1$, the purity obstruction gives the upper bound one, and Proposition 14 supplies the matching boundary value, with no finite attainment. For $\alpha > 1$, equality in (32) forces the optimal canonical effect to be $u = 0$; hence the original positive filter is proportional to W. Formula (101) at $u = 0$ gives $1 - \|a\|$.

Substitution of (34) proves (80). The common numerator satisfies $J_\rho = 4\operatorname{tr}(\rho\widetilde{\rho}) \ge 0$. Swapping the parties therefore changes only the positive denominator $1 - \|a\|^2$ to $1 - \|b\|^2$, which proves the weak ordering. If both ratios are at most one, the two optima are equal on the plateau even when the local Bloch lengths differ. □

In particular, at $\alpha = 1$ finite attainment of the local bound is equivalent to $\operatorname{rank}\rho \le 2$: the theorem proves sufficiency, and Proposition 14 proves necessity.

For an explicit equality case of rank three, take

$$\rho_{\text{threshold}} = \frac{1}{8}\begin{pmatrix} 4 & 0 & 0 & 2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 2 & 0 & 0 & 3 \end{pmatrix}. \tag{82}$$

Here $D = \operatorname{diag}(1/2, -1/2, 3/4)$, $c = (0, 0, 1/4)$, and $\alpha = 1$. The eigenvalues of H_0 are $0, 3/8, 1/2$, and its kernel is spanned by $(i, 1, 0)^T$, which is isotropic for $z^T z$. Thus $B_A^* = 1$, but no invertible filter attains it. The density-matrix kernel is generated by the product vector $|01\rangle$, which also verifies $E = 1$ directly.

12 Exact examples and checks

12.1 Whitening alone can miss activation

Consider the physical rank-three state

$$\rho_1 = \frac{1}{40}\begin{pmatrix} 20 & 0 & 0 & 14 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 7 & 0 \\ 14 & 0 & 0 & 13 \end{pmatrix}. \tag{83}$$

Its Alice marginal is already maximally mixed, and

$$c = (0, 0, 7/20), \qquad D = \mathrm{diag}(7/10, -7/10, 13/20), \qquad B(\rho_1)^2 = 49/50. \tag{84}$$

The matrix M has eigenvalues $419/1200$ (twice) and $169/400$. The first eigenspace is admissible with $\|v\| = 13/21$. The boundary radius is $E = 1$. The theorem gives

$$\boxed{(B_A^*(\rho_1))^2 = 79/75 > 1.} \tag{85}$$

One optimal effect direction is $u = (0, 0, 13/21)$, realized by

$$F_A = \mathrm{diag}(1, 2/\sqrt{17}), \qquad \Pr(\text{success}) = 21/34. \tag{86}$$

The filtered correlation singular values are $13/15, 2\sqrt{17}/15, 2\sqrt{17}/15$. This independently confirms the value $79/75$. The Bob-only value for the same state is $(B_B^*)^2 = 292/247$.

12.2 A full-rank state activated only by Alice

Let

$$\rho_2 = \frac{1}{500}\begin{pmatrix} 441 & 0 & 0 & 56 \\ 0 & 49 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 56 & 0 & 0 & 9 \end{pmatrix}. \tag{87}$$

Its Bloch data are $a = (0, 0, 24/25)$, $b = (0, 0, 96/125)$, and $T = \mathrm{diag}(28/125, -28/125, 4/5)$. Directly,

$$B(\rho_2)^2 = 10784/15625 < 1. \tag{88}$$

The Alice filter $\mathrm{diag}(1/7, 1)$ succeeds with probability $1/25$ and returns the isotropic Bell mixture with visibility $4/5$. The unital case of the theorem proves optimality:

$$\boxed{B_A^*(\rho_2) = 4\sqrt{2}/5 > 1.} \tag{89}$$

On whitening Bob, the other marginal and correlation matrix are

$$c = (0, 0, 5400/6409), \qquad D = \mathrm{diag}(28/\sqrt{6409}, -28/\sqrt{6409}, 980/6409). \tag{90}$$

Here $K \prec 0$, whence $CK^{-1}C \succeq 0$ and $M \succeq A$. Every interior value is at most $\mathrm{tr}\, A < 0.27$. The boundary quadratic has its vertex at $5292000/4064256 > 1$, so its maximum on $[-1, 1]$ is at the positive z pole and gives

$$\boxed{B_B^*(\rho_2) = 220/221 < 1.} \tag{91}$$

This value has no invertible-filter maximizer: every interior level is strictly smaller. This one exact state illustrates unilateral asymmetry, strict nonactivation on one side, and the necessity of a supremum in Equation (9).

12.3 Consistency and degeneracy checks

For $c = 0$, $C = 0$ and the formula reduces to $(B_A^*)^2 = \operatorname{tr} A - \lambda_{\min}(A)$, the sum of the two largest squared singular values of D. This recovers the unital-channel result [8]. For the canonical product state $I/2 \otimes \rho_B$, it gives $B_A^* = \|c\|$; if $c \neq 0$, only rank-one Alice filters attain that value in the closed domain. For $I_4/4$, the answer is zero. For every pure entangled state the answer is $\sqrt{2}$.

The exceptional level $\lambda = s$ is not merely a boundary phenomenon. One physical example is

$$D = \operatorname{diag}(\sqrt{3/10}, -1/2, 1/\sqrt{5}), \qquad c = (0, 0, \sqrt{q}), \qquad q = \frac{15 - \sqrt{105}}{40}. \tag{92}$$

It has a nonisotropic complex kernel at the exceptional level and a feasible interior value $3/4 - q$ that exceeds the boundary value. The supplied independent check treats this repeated-eigenvalue branch explicitly.

13 An exact phase-covariant activation region

Consider the three-parameter canonical family

$$D = \operatorname{diag}(t, -t, r), \qquad c = (0, 0, c_0), \qquad x = t^2, \tag{93}$$

with

$$0 \le r \le 1, \qquad 0 \le c_0 \le 1 - r, \qquad 4x \le (1 + r)^2 - c_0^2. \tag{94}$$

These conditions are necessary and sufficient for positivity in this parameter range: the two uncoupled diagonal entries are $(1 - c_0 - r)/4$ and $(1 + c_0 - r)/4$, and the remaining 2×2 block has nonnegative determinant precisely under the last inequality. The family includes amplitude damping and the loss channels studied in [8].

Theorem 16 (Exact axial activation)**.** *For the physical family* (93)*,*

$$\boxed{\begin{aligned} B_A^* > 1 \quad \Longleftrightarrow \quad & x > \tfrac{1}{2} \\ & \text{or} \quad \left[x \le \tfrac{1}{2},\ r > 1 - x,\ c_0^2 > \frac{x(1 - x - r^2)}{1 - x} \right]. \end{aligned}} \tag{95}$$

Whenever $\alpha = 2x + r^2 - c_0^2 > 1$, its exact value is

$$(B_A^*)^2 = \max\left\{ 2x,\ x + \frac{x r^2}{x - c_0^2} \right\}. \tag{96}$$

The canonical state exhibits hidden activation exactly in the region

$$0 < x \le \tfrac{1}{2}, \qquad 1 - x < r \le \sqrt{1 - x}, \qquad c_0^2 > \frac{x(1 - x - r^2)}{1 - x}, \tag{97}$$

together with (94)*. In this hidden region an optimal effect has*

$$u = (0, 0, u_3), \qquad u_3 = \frac{r c_0}{x - c_0^2}, \qquad p_{\max} = \frac{x - c_0^2}{x - c_0^2 + r c_0}. \tag{98}$$

Proof. The vector d is $(0, 0, r c_0)$, and K is diagonal. In the guarded region the eigenvalues of M are

$$x - \frac{r^2 c_0^2}{x - c_0^2}, \quad x - \frac{r^2 c_0^2}{x - c_0^2}, \quad r^2.$$

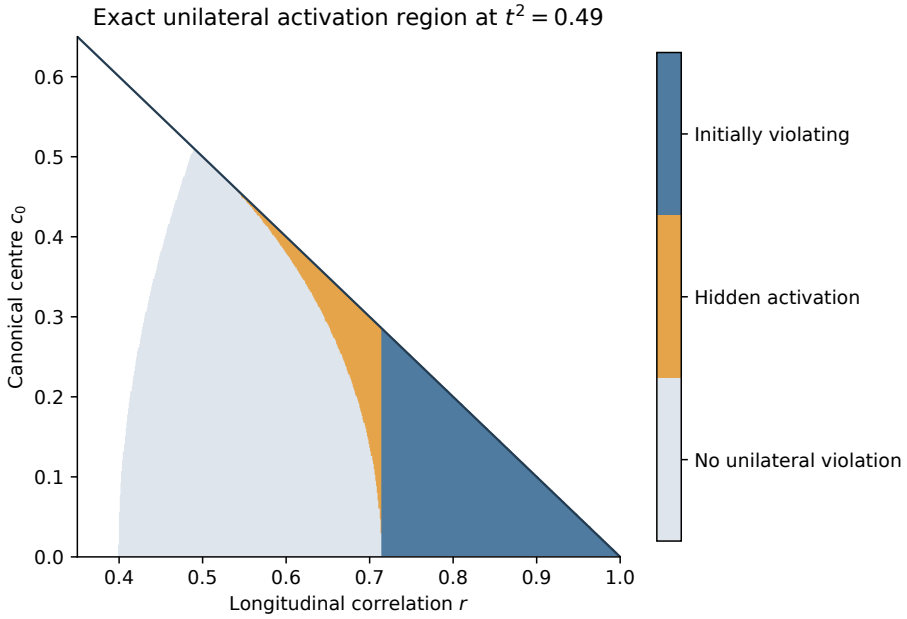


Figure 1: The exact axial activation region at $x = t^2 = 0.49$. The physical region is separated into states with no unilateral violation, states with hidden activation, and states that already violate CHSH. The boundaries are given by (94) and (95).

Theorem 3 gives (96). If $x > 1/2$, the original state already has $B^2 \geq 2x > 1$. Suppose $x \leq 1/2$, and set

$$L = \frac{x(1-x-r^2)}{1-x}, \qquad U = 2x + r^2 - 1.$$

The second branch in (96) exceeds one exactly when $c_0^2 > L$; the purity guard is $c_0^2 < U$. Their possible overlap satisfies

$$U - L = \frac{r^2 - (1-x)^2}{1-x} > 0 \quad \Longleftrightarrow \quad r > 1 - x.$$

Conversely, if $r > 1-x$, physicality implies $c_0^2 \leq (1-r)^2 < U$, because $U-(1-r)^2 = 2(x+r-1) > 0$. This proves the exact activation condition without retaining a separate guard.

Initially $B^2 = x+\max(x, r^2)$, so initial nonviolation means $x \leq 1/2$ and $x+r^2 \leq 1$. Combining these with (95) gives (97). In this region the transverse eigenspace supplies the winning branch; substituting $n = e_1$ in (36) gives $u_3 = rc_0/(x - c_0^2)$. Its norm is strictly below one by automatic feasibility. Rescaling its Kraus operator to its largest admissible size gives $p_{\max} = 1/(1+u_3)$. □

Corollary 17 (The quasi-distillable family in the target paper)**.** *For*

$$\rho_p = p|\Psi^-\rangle\langle\Psi^-| + (1-p)|00\rangle\langle 00|, \qquad 0 \leq p \leq 1,$$

one has

$$\boxed{(B_A^*(\rho_p))^2 = (B_B^*(\rho_p))^2 = \max\left\{1, \frac{2p}{2-p}\right\}.} \tag{99}$$

For $0 < p < 2/3$, the value one is a supremum attained only by a rank-one limit. At and above $p = 2/3$, an invertible whitening filter attains the optimum. At $p = 0$, the state is already a pure product.

Proof. For $p > 0$, Alice whitening and a local rotation produce the amplitude damping form $x = r$, $c_0 = 1 - r$, with $r = p/(2-p)$. The density matrix has rank two, so its boundary value is

one. For $r < 1/2$, $\alpha = 4r - 1 < 1$, and Theorem 13 gives exactly one with no finite attainment. At $r = 1/2$, the canonical state already has $B^2 = 2r = 1$. For $r > 1/2$, formula (96) has maximum $2r$: the inequality for the other branch reduces to $(2r-1)(r-1) \le 0$. Symmetry of ρ_p exchanges Alice and Bob. □

This gives a conjecture-free, sharp interval of locally inaccessible hidden CHSH violation for this family: $0 < p \le 2/3$. For comparison, both steering-ellipsoid centres of ρ_p have length $2(1-p)/(2-p)$, which is at least $1/2$ exactly when $p \le 2/3$. Result 1 of [1] derives the absence of one-sided violation for centre lengths above $1/2$ from Conjecture 1 of [14], and at centre length $1/2$ that conjecture also gives $B \le 1$; the Omega Institute's public record of a claimed proof of Conjecture 1 [16] therefore already makes the inaccessibility of $0 < p \le 2/3$ unconditional, and no priority is claimed for it. What the corollary adds is the exact value (99) and the sharpness of the endpoint: for every $p > 2/3$ whitening alone gives a one-sided violation. Indeed, the physical bilateral filters $F_\varepsilon = \mathrm{diag}(\varepsilon, 1)$ on both parties, with $0 < \varepsilon \le 1$, produce the same family with parameter and success probability

$$q = \frac{p}{p + (1-p)\varepsilon^2}, \qquad p_{\mathrm{succ}} = p\varepsilon^2 + (1-p)\varepsilon^4.$$

For every $p > 0$, $q \to 1$ as $\varepsilon \to 0$, so the bilateral CHSH supremum is $\sqrt{2}$. Directly, $B(\rho_q)^2 = \max\{2q^2, q^2 + (1-2q)^2\}$. For $0 < p < 1$, the explicit sufficient and necessary condition $q > 1/\sqrt{2}$ is obtained by choosing $\varepsilon^2 < p(\sqrt{2}-1)/(1-p)$. Thus every $0 < p \le 2/3$ permits a finite-success bilateral violation although neither unilateral filter can violate. For the exact example $p = 3/5$ and $\varepsilon = 1/2$, the initial squared CHSH value is $18/25$, both unilateral suprema have square one, and the bilateral filtered value has square $72/49$, at success probability $7/40$. A separate exact rational certificate is included.

The amplitude-damping threshold $r > 1/2$ and the loss-channel golden-ratio threshold are established special cases in Appendix B of [8]; they are not being claimed as new here. For example, setting $x = r^2$ and $c_0 = 1 - r$ in (95) gives $r > (\sqrt{5}-1)/2$. The theorem supplies one criterion for the full three-parameter physical family and its exact hidden-activation region.

14 Success probabilities and an exact constrained dual

An optimum conditional CHSH value does not determine a unique success probability, because a Kraus operator may be scaled down arbitrarily. Throughout this section, $p_{\max}$ denotes the largest probability compatible with $F^\dagger F \le I$ for the specified conditional filter.

Proposition 18 (Exact maximal success probability)**.** *For a canonical effect $I + u \cdot \sigma$,*

$$p_{\max}(0, u) = \frac{1}{1 + \|u\|}. \tag{100}$$

For the original mixed Alice marginal with Bloch vector a, the whitening-and-filtering branch $F = \sqrt{I + u \cdot \sigma}\,(I + a \cdot \sigma)^{-1/2}$ has

$$\boxed{p_{\max}(a, u) = \frac{1 - \|a\|^2}{1 - a \cdot u + \sqrt{(1 - a \cdot u)^2 - (1 - \|a\|^2)(1 - \|u\|^2)}}.} \tag{101}$$

Proof. The unscaled branch has success normalization one. If $E = F^\dagger F$, its largest allowed scalar rescaling has squared size $1/\lambda_{\max}(E)$, and hence this is its maximum probability. For a canonical input the effect eigenvalues are $1 \pm \|u\|$. For the original state,

$$\mathrm{tr}\, E = \frac{2(1 - a \cdot u)}{1 - \|a\|^2}, \qquad \det E = \frac{1 - \|u\|^2}{1 - \|a\|^2}.$$

The quadratic formula for the larger eigenvalue gives (101). □

The next theorem concerns the probability constraint on the *canonical input*. This restriction is substantive: whitening a general input changes success probabilities, and (101), not (100), must be used when imposing a probability constraint on the original state.

Theorem 19 (One-dimensional convex dual for a success constraint)**.** *Suppose $H_0 = K + iC \succ 0$, which holds in particular when $\alpha > 1$. Require a canonical filter to have maximum success probability at least $p_0 \in [1/2, 1]$, and put $R = p_0^{-1} - 1 \in [0, 1]$. Let S_R be the largest squared CHSH value under that constraint. Then*

$$\boxed{S_R = \inf_{\tau \geq 0} \left[\operatorname{tr} A + \tau R^2 - \lambda_{\min}\Big(A + C(K + \tau I)^{-1} C\Big) \right].} \tag{102}$$

The function minimized is convex in τ. For $R > 0$ it has a finite minimizer, and a corresponding optimal filter can be reconstructed from its minimum eigenspace. The value is nondecreasing and concave as a function of R^2. It reaches the unconstrained plateau at

$$R_*^2 = \min_{\substack{\|n\|=1 \\ Mn = \lambda_{\min}(M) n}} \left\| K^{-1} C n \right\|^2. \tag{103}$$

At $R = 0$, $S_0 = \operatorname{tr} A - \lambda_{\min}(A)$ is the original canonical CHSH value.

Proof. The probability constraint is exactly $\|u\| \leq R$. The Lorentz reduction in the proof of the main theorem respects it. Indeed its replacement filter has squared effect norm $1 - 1/\Delta$, whereas the original filter has squared effect norm $1 - 1/w_0^2$, and $\Delta \leq w_0^2$. Thus replacement decreases the norm and cannot lower the chosen correlation-plane value. Conversely each reduced feasible pair gives $u = -n \times v$ with $\|u\| = \|v\|$. It follows that S_R is exactly the maximum of $f(n, v)$ subject to $\|n\| = 1$, $n \cdot v = 0$, and $\|v\| \leq R$.

For $\tau \geq 0$, write $P_\tau = K + \tau I$ and $M_\tau = A + C P_\tau^{-1} C$. Completing the square gives, for every feasible pair,

$$\begin{aligned} f(n, v) = {} & \operatorname{tr} A - n^T M_\tau n + \tau \|v\|^2 \\ & - (v + P_\tau^{-1} C n)^T P_\tau (v + P_\tau^{-1} C n) \leq \operatorname{tr} A + \tau R^2 - \lambda_{\min}(M_\tau). \end{aligned} \tag{104}$$

This proves weak duality. The expression on the right is also the maximum, over $\|n\| = 1$ and arbitrary real v, of $f(n, v) + \tau(R^2 - \|v\|^2)$, with the orthogonality constraint temporarily omitted. It is therefore a supremum of affine functions of τ, and is convex. For $R > 0$ it tends to infinity as $\tau \to \infty$, so it has a minimizer.

Let N_τ have orthonormal columns spanning the minimum eigenspace of M_τ. The possible squared norms of $v = -P_\tau^{-1} C n$, for unit n in that eigenspace, form the whole Rayleigh interval

$$[r_-^2(\tau), r_+^2(\tau)] = \left[\lambda_{\min}(N_\tau^T C^T P_\tau^{-2} C N_\tau), \lambda_{\max}(N_\tau^T C^T P_\tau^{-2} C N_\tau) \right]. \tag{105}$$

Here $M_\tau' = -C P_\tau^{-2} C = C^T P_\tau^{-2} C$. The left and right derivatives of the convex dual function are $R^2 - r_+^2(\tau)$ and $R^2 - r_-^2(\tau)$, respectively. At a positive minimizing τ, they bracket zero, so choose a unit minimum eigenvector whose associated v has norm exactly R. At a minimizer $\tau = 0$, the right derivative is nonnegative, so choose one with $\|v\| \leq R$.

It remains to check the omitted orthogonality condition. The positive matrix $H_0 + \tau I = P_\tau + iC$ has Schur complement

$$P_\tau + C P_\tau^{-1} C = M_\tau + (\tau - s) I \succ 0.$$

The paired eigenvector equations therefore force $(\lambda_{\min}(M_\tau) + \tau - s) n \cdot v = 0$ with a strictly positive coefficient, so $n \cdot v = 0$. The selected pair is feasible and gives equality in (104), proving strong duality. This argument uses the entire eigenspace and therefore includes eigenvalue crossings.

The infimum of affine functions of R^2 is concave; monotonicity also follows directly from nesting of the feasible balls. At $\tau = 0$, the same one-sided derivative test is precisely $R^2 \geq R_*^2$, proving the plateau assertion. For $R = 0$, the primal domain forces $v = 0$, and the dual tends to the same value as $\tau \to \infty$. □

Equation (102) is an exact convex scalar representation. In general it is not asserted to be a formula in radicals. The axial family does permit a complete piecewise expression, including an interval where the best filter breaks the apparent axial symmetry.

Theorem 20 (Closed axial frontier and tilted optimal filters)**.** *For (93), suppose $H_0 \succ 0$, and put*

$$k = x - c_0^2, \qquad g = rc_0 > 0, \qquad R_* = \frac{g}{k} < 1.$$

If $x \le r^2$, then

$$S_R = \begin{cases} x + r^2 + 2gR - kR^2, & 0 \le R \le R_*, \\ x + r^2 + g^2/k, & R \ge R_*. \end{cases} \tag{106}$$

If $x > r^2$ and $g^2 \le k(x - r^2)$, then $S_R = 2x$ for every R. In the remaining case, define

$$\tau_c = \frac{g^2}{x - r^2} - k > 0, \qquad R_c = \frac{x - r^2}{g}, \qquad 0 < R_c < R_*.$$

Then

$$\boxed{S_R = \begin{cases} 2x + \tau_c R^2, & 0 \le R \le R_c, \\ x + r^2 + 2gR - kR^2, & R_c \le R \le R_*, \\ x + r^2 + g^2/k, & R \ge R_*. \end{cases}} \tag{107}$$

For $0 < R < R_c$, one optimal effect is

$$u = \left(-R\sqrt{1 - (R/R_c)^2},\ 0,\ R^2/R_c \right), \tag{108}$$

up to rotations about the symmetry axis. Every optimal effect in this open interval has a nonzero component perpendicular to that axis. For $R_c \le R \le R_*$, an optimal effect is $u = (0, 0, R)$; above the plateau, replace R by R_*.

Proof. Positivity of H_0 is equivalent here to $k > g$ and $r^2 > c_0^2$. The eigenvalues of M_τ are $x - g^2/(k + \tau)$, twice, and r^2. Consequently the dual objective is the *upper* envelope

$$\max \left\{ x + r^2 + \tau R^2 + \frac{g^2}{k + \tau},\ 2x + \tau R^2 \right\}. \tag{109}$$

If $x \le r^2$, the first branch is always active. Its minimizing parameter for $R > 0$ is $\tau = \max(g/R - k, 0)$, yielding (106); at $R = 0$ the infimum is obtained as $\tau \to \infty$. If $x > r^2$ but $\tau_c \le 0$, the second branch is always active, and $\tau = 0$ gives the constant value $2x$.

In the remaining case the first branch applies for $0 \le \tau \le \tau_c$, and the second for $\tau \ge \tau_c$. For $0 < R < R_c$ their left and right slopes at the crossing have opposite signs, so the minimum is at τ_c. At $R = 0$, every $\tau \ge \tau_c$ has the minimum value $2x$. For $R_c \le R \le R_*$ it is at $\tau = g/R - k$; for $R \ge R_*$ it is at zero. Substitution proves all three pieces.

At τ_c, the minimum eigenspace is all of $\mathbb{R}^3$. Set $\theta = (R/R_c)^2$ and choose

$$n = (\sqrt{\theta}, 0, \sqrt{1 - \theta}), \qquad v = (0, -R_c\sqrt{\theta}, 0).$$

These vectors satisfy $\|n\| = 1$, $n \cdot v = 0$, and $\|v\| = R$, and give (108). Moreover every equality pair must have $v = -(K + \tau_c I)^{-1} C n$ and $\|v\| = R$. It follows that the transverse squared norm of n is θ, so $\|u_\perp\| = R\sqrt{1 - \theta} > 0$. Thus a restriction to axial effects loses the true optimum throughout this interval. □

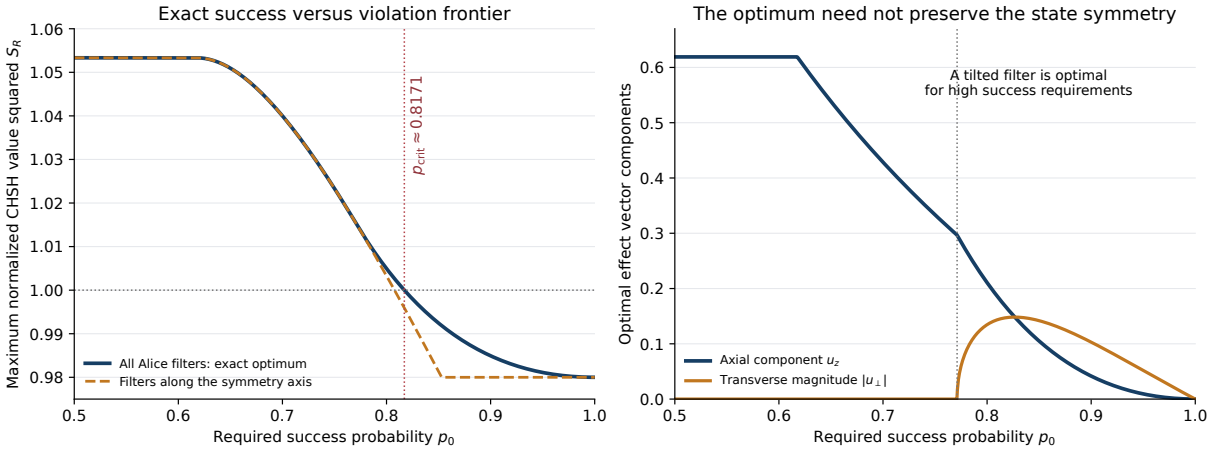


Figure 2: The exact success frontier for ρ_1. Left: the unrestricted optimum (110) compared with filters constrained to the symmetry axis. Right: the axial and transverse components of an optimal effect. A nonzero transverse component is necessary throughout the high-success interval corresponding to $0 < R < R_c$.

14.1 An exact probability threshold inside the tilted regime

Return to the canonical activation example $\rho_1 = \frac{1}{40}\begin{pmatrix} 20 & 0 & 0 & 14 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 7 & 0 \\ 14 & 0 & 0 & 13 \end{pmatrix}$ from the earlier examples. Here

$$x = \frac{49}{100}, \quad r = \frac{13}{20}, \quad c_0 = \frac{7}{20}, \quad k = \frac{147}{400}, \quad g = \frac{91}{400}.$$

The exact breakpoints and scalar dual parameter are

$$R_c = \frac{27}{91}, \qquad R_* = \frac{13}{21}, \qquad \tau_c = \frac{539}{1350}.$$

Thus

$$S_R = \begin{cases} \dfrac{49}{50} + \dfrac{539}{1350}R^2, & 0 \le R \le 27/91, \\ \dfrac{73}{80} + \dfrac{91}{200}R - \dfrac{147}{400}R^2, & 27/91 \le R \le 13/21, \\ \dfrac{79}{75}, & R \ge 13/21. \end{cases} \tag{110}$$

The constrained state violates CHSH exactly when

$$R > \sqrt{\frac{27}{539}} \quad \Longleftrightarrow \quad p_0 < \frac{1}{1 + \sqrt{27/539}}. \tag{111}$$

Since $\sqrt{27/539} < 27/91$, the first onset of violation occurs strictly inside the tilted-filter regime. This is a concrete obstruction to optimizing the success tradeoff by assuming that an axially symmetric input has an axial optimal filter.

15 Success constraints on the original input state

The earlier scalar dual constrains the success probability on a canonical input. Whitening is itself probabilistic, so that dual does not directly solve the tradeoff on the original state. This section gives the correct feasible geometry, an exact algebraic optimization for every input, and closed frontiers for several noncanonical families.

Let the original Alice marginal have Bloch vector a, with $r = \|a\| < 1$, and put $W = (I + a \cdot \sigma)^{-1/2}$. Write the canonical data as (c, D) and set $Q = D^T D$. For $0 \le p \le 1$, define

$$S_\rho(p) = \sup_{\substack{F \text{ invertible, } F^\dagger F \preceq I \\ \mathrm{tr}[(F^\dagger F \otimes I)\rho] \ge p}} B(\rho_F)^2.$$

Thus p is a required success probability, rather than the probability of a particular arbitrarily rescaled Kraus operator.

15.1 Two balls, rather than a centered-radius constraint

Theorem 21 (Exact original-input success lens)**.** *A canonical effect vector u, $\|u\| \le 1$, can be implemented on the original state with success probability at least p if and only if*

$$\boxed{\|a - pu\| \le 1 - p.} \tag{112}$$

Consequently its feasible set is

$$\mathcal{L}_p(a) = \{u : \|u\| \le 1,\ \|a - pu\| \le 1 - p\}. \tag{113}$$

For $p > 0$, the second condition is the ball centered at a/p of radius $(1-p)/p$. The constraint is inactive when $p \le (1-r)/2$. That second ball lies inside the unit ball when $p \ge (1+r)/2$. At $p = 1$ the lens is the singleton $\{a\}$.

The smallest and largest feasible norms are

$$r_-(p) = \max\left\{0, 1 - \frac{1-r}{p}\right\}, \qquad r_+(p) = \min\left\{1, \frac{1+r}{p} - 1\right\}, \tag{114}$$

with $r_-(0) = 0$ and $r_+(0) = 1$.

Proof. The unscaled combined filter is $F_u = \sqrt{I + u \cdot \sigma}\, W$, and its success normalization on ρ is one. Its largest physical success probability is $1/\lambda_{\max}(E_u)$, where $E_u = W(I + u \cdot \sigma)W$. Hence success at least p is equivalent to $pE_u \preceq I$. Congruence by W^{-1} turns this into

$$p(I + u \cdot \sigma) \preceq I + a \cdot \sigma.$$

The difference has eigenvalues $1 - p \pm \|a - pu\|$, proving (112). The two ball-containment conditions follow by comparing their center distance r/p and radii $1, (1-p)/p$.

The nearest point of the second ball to the origin has distance $\max(0, [r - (1-p)]/p) = r_-(p)$, and it lies in the unit ball since $r_-(p) \le r < 1$. The farthest feasible distance is the smaller of one and the second ball's farthest distance $(r + 1 - p)/p$. Both extrema are obtained on the line through a; if $a = 0$, any radial direction may be used for the maximum. □

In particular, whitening $u = 0$ is allowed exactly when $p \le 1 - r$, which is a different threshold from the condition that *every* canonical filter is allowed.

Lemma 22 (Density of invertible filters and attainment above one)**.** *Maximizing a continuous filtered score over the closed lens gives the same value as the invertible-filter supremum. Every constrained optimum with $S_\rho(p) > 1$ is attained by an invertible filter.*

Proof. For $p < 1$, the point $u = a$ is strictly inside both defining balls: $\|a\| < 1$ and $\|a - pa\| = (1-p)r < 1 - p$. For any feasible u, the points $u_t = (1-t)u + ta$, $0 < t \le 1$, remain feasible by convexity and satisfy $\|u_t\| < 1$. They converge to u, so invertible effects are dense. At $p = 1$, the only feasible point is already the interior point a. Finally, $\|u\| = 1$ gives a rank-one Alice filter and therefore a product output, whose normalized CHSH value is at most one. A compact maximizer with value greater than one must consequently have $\|u\| < 1$. □

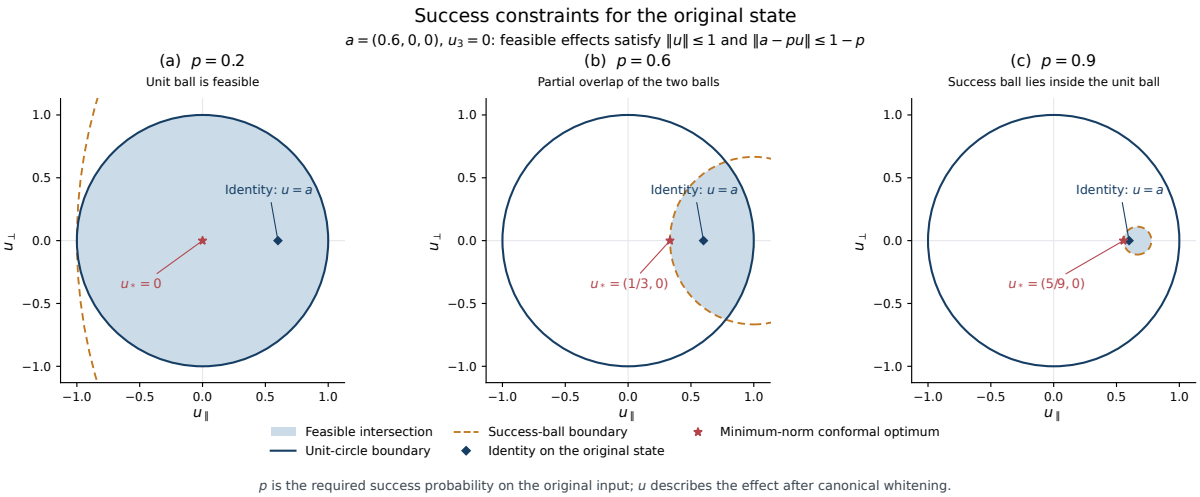


Figure 3: Planar sections of the exact original-input success lens at $\|a\| = 0.6$. As the required probability increases, the feasible region changes from the entire effect ball to a lens and then to a smaller ball. The identity filter on the original state corresponds to $u = a$ and remains feasible. The closest point to the origin is the exact optimum for the conformal family of Theorem 27.

15.2 The correct quadratic Gram matrix

Theorem 23 (An exact three-variable constrained optimization)**.** *Define the polynomial symmetric matrix*

$$\boxed{G(u) = (1 - \|u\|^2)(Q - cc^T) + (c + D^T u)(c + D^T u)^T.} \tag{115}$$

Then, for every canonical effect including $\|u\| = 1$, $G(u) = T_u^T T_u$. *In particular*

$$\boxed{S_\rho(p) = \max_{u \in \mathcal{L}_p(a)} [\operatorname{tr} G(u) - \lambda_{\min}(G(u))].} \tag{116}$$

Equivalently, for each unit Bob-side normal n, *put* $P_n = I - nn^T$,

$$k_n = \operatorname{tr}[P_n(Q - cc^T)], \qquad h_n = DP_n c, \qquad J_n = k_n I - DP_n D^T.$$

The objective associated with that measurement plane is the quadratic

$$f_n(u) = \operatorname{tr}(P_n Q) + 2h_n^T u - u^T J_n u, \qquad S_\rho(p) = \max_{\|n\|=1} \max_{u \in \mathcal{L}_p(a)} f_n(u). \tag{117}$$

Proof. Set $q = \sqrt{1 - \|u\|^2}$ and $S_u = qI + uu^T/(1 + q)$. The filtered correlation matrix is $T_u = S_u D + uc^T$. The identities $S_u u = u$ and $S_u^2 = (1 - \|u\|^2)I + uu^T$ give

$$T_u^T T_u = (1 - \|u\|^2)Q + (D^T u)(D^T u)^T + (D^T u)c^T + c(D^T u)^T + \|u\|^2 cc^T,$$

which is (115). This calculation remains valid at the rank-one boundary. The Horodecki formula gives (116); compactness and Lemma 22 justify the maximum. Taking $\operatorname{tr}(P_n G(u))$ and expanding proves (117). □

The Bob-side normal in (117) is unrestricted. It must not be constrained to be perpendicular to u. The Lorentz replacement in the unconstrained proof can change the effect vector toward the origin, which need not preserve a ball centered at a/p.

Remark 24 (An exact counterexample to the naive shifted reduction). Take the full-rank original state

$$\rho = \frac{1}{200}\begin{pmatrix} 116 & 0 & 0 & 56 \\ 0 & 44 & 8 & 0 \\ 0 & 8 & 11 & 0 \\ 56 & 0 & 0 & 29 \end{pmatrix}. \tag{118}$$

The two block determinants before division by 200^2 are 228 and 420, so positivity is strict. Here $a = (0, 0, 3/5)$, and the canonical data are $c = 0$, $D = \mathrm{diag}(4/5, -3/5, 9/20)$. Thus $K = DD^T \succ 0$ and $\alpha = 481/400 > 1$. At $p = 1$, only $u = a$ is feasible, and the true score is $S_\rho(1) = 16/25$. If one instead imposes $u = -n \times v = a$ in the earlier reduced objective, one forces $n \perp a$ and obtains only

$$(16/25)^2 + (9/20)^2 = \frac{6121}{10000} < \frac{16}{25}.$$

The failure therefore occurs even inside the positive-Gram regime.

15.3 A rational description using the ellipsoid and original marginals

Theorem 25 (The full frontier needs no canonical orientation)**.** *Let b be the original Bob Bloch vector, $g = b - c = D^T a$, and $Q = D^T D$. If $Q \succ 0$, set $w = D^T u$. Then the exact frontier is obtained by maximizing the top-two eigenvalue sum of*

$$\mathcal{G}(w) = (1 - w^T Q^{-1} w)(Q - cc^T) + (c + w)(c + w)^T \tag{119}$$

under the two ellipsoid constraints

$$w^T Q^{-1} w \le 1, \qquad (g - pw)^T Q^{-1} (g - pw) \le (1 - p)^2. \tag{120}$$

All coefficients are rational functions of the original Bloch data.

The same statement extends to singular Q as follows. Write $Q^\dagger$ for its Moore–Penrose inverse, restrict $w \in \mathrm{ran}\, Q$, and put

$$\eta = \|a\|^2 - g^T Q^\dagger g \ge 0.$$

If $\mathrm{rank}\, Q < 3$ *and $\eta > 0$, introduce $k \ge 0$ and set*

$$q = w^T Q^\dagger w + k^2/\eta, \tag{121}$$

$$\ell = (g - pw)^T Q^\dagger (g - pw) + \eta - 2pk + p^2 k^2/\eta. \tag{122}$$

If $\mathrm{rank}\, Q < 3$ *and $\eta = 0$, introduce $t \ge 0$ and instead set*

$$q = w^T Q^\dagger w + t, \qquad \ell = (g - pw)^T Q^\dagger (g - pw) + p^2 t. \tag{123}$$

In both cases maximize the top-two eigenvalue sum of

$$(1 - q)(Q - cc^T) + (c + w)(c + w)^T \tag{124}$$

under $q \le 1$ and $\ell \le (1 - p)^2$. This is exact, including all rank-zero, rank-one, and rank-two shapes. After choosing coordinates in $\mathrm{ran}\, Q$*, it uses at most three real variables and has rational coefficients on each rank stratum.*

Proof. When D is invertible, $u = D^{-T} w$ and $a = D^{-T} g$, proving (120) and (119). The equality $g = b - c$ follows by applying the effect $u = a$, which undoes whitening and restores the original Bob marginal. Alternatively it follows directly from $c = (b - T^T a)/(1 - \|a\|^2)$.

For the singular case decompose orthogonally

$$u = DQ^\dagger w + u_0, \qquad a = DQ^\dagger g + a_0, \qquad u_0, a_0 \in \ker D^T.$$

Then $\|a_0\|^2 = \eta$ and $\|u\|^2 = w^T Q^\dagger w + \|u_0\|^2$. The Gram matrix depends on u_0 only through its squared norm. For a fixed norm, the smallest possible value of $\|a_0 - pu_0\|^2$ is obtained by aligning u_0 with a_0. If $\eta > 0$, write $u_0 = (k/\eta)a_0$ with $k \geq 0$. Its norm and distance are exactly the extra terms in (122). If $\eta = 0$, choose any null-space vector of squared norm t, giving (123). This alignment preserves the score and can only improve success feasibility; every reduced feasible point reconstructs a real filter.

The raw formulas for c, Q are rational, as in (73). The pseudoinverse is rational on each rank stratum. Explicitly, with $e_1 = \operatorname{tr} Q$ and $e_2 = ((\operatorname{tr} Q)^2 - \operatorname{tr} Q^2)/2$,

$$Q^\dagger = \begin{cases} 0, & \operatorname{rank} Q = 0, \\ Q/e_1^2, & \operatorname{rank} Q = 1, \\ [(e_1^2 - e_2)Q - e_1 Q^2]/e_2^2, & \operatorname{rank} Q = 2, \\ \operatorname{adj}(Q)/\det Q, & \operatorname{rank} Q = 3. \end{cases}$$

Each formula follows by evaluation on the nonzero eigenvalues. For rank below three, there are $\operatorname{rank} Q$ range coordinates and one additional scalar, proving the dimension bound. □

This theorem says precisely which extra data the original-input tradeoff needs: the steering ellipsoid (Q, c), the original Bob point b, and the length $\|a\|$. In the nonsingular case the latter is already encoded by $g^T Q^{-1} g$. The conditional Bob Bloch vector is $c + w$; the second ellipsoid constraint is the condition that the complementary outcome can also have a conditional Bob vector in the same ellipsoid.

15.4 Exact algebraic decisions and verifiable upper certificates

Proposition 26 (Degree-six decision and a complete certificate hierarchy)**.** *For a proposed squared-score threshold γ, define*

$$H_\gamma(u) = \gamma I - \operatorname{tr} G(u) I + G(u), \quad g_1(u) = 1 - \|u\|^2, \quad g_2(u) = (1-p)^2 - \|a - pu\|^2 .$$

Then $S_\rho(p) > \gamma$ exactly when there is a real u with $g_1, g_2 \geq 0$ for which at least one of

$$\operatorname{tr} H_\gamma, \qquad \frac{(\operatorname{tr} H_\gamma)^2 - \operatorname{tr}(H_\gamma^2)}{2}, \qquad \det H_\gamma \tag{125}$$

is negative. These are polynomials of degree at most $2, 4, 6$ in the three coordinates of u. Thus real quantifier elimination gives an exact algebraic decision for algebraic input data.

An exact polynomial identity

$$H_\gamma = \Sigma_0 + g_1 \Sigma_1 + g_2 \Sigma_2, \qquad \Sigma_j = \sum_\ell V_{j\ell}^T V_{j\ell}, \tag{126}$$

is a sufficient upper certificate $S_\rho(p) \leq \gamma$. For every strict upper threshold $\gamma > S_\rho(p)$, such a matrix sum-of-squares certificate exists at some finite degree.

Proof. The largest eigenvalue of $\operatorname{tr} G\, I - G$ is the sum of the two largest eigenvalues of G. Thus $H_\gamma \succeq 0$ is exactly the pointwise upper-score condition. The three elementary symmetric coefficient test for a real symmetric 3×3 matrix proves (125); the degree bounds follow because G is quadratic in u.

Every right-hand term in (126) is positive semidefinite on the lens, proving sufficiency by direct evaluation. For completeness, if $\gamma > S_\rho(p)$, then H_γ is uniformly positive definite on the compact lens. The matrix Positivstellensatz of Hol–Scherer [13] applies: the quadratic module is Archimedean because it already contains $1 - \|u\|^2$. It supplies the displayed representation. No fixed degree is asserted, and at equality $\gamma = S_\rho(p)$ finite-degree attainment is not guaranteed. □

Thus one may search for upper certificates by semidefinite programming, but floating-point feasibility alone is not a certificate: the polynomial identity and the positive semidefinite Gram matrices must be checked exactly or with rigorous error bounds. Likewise, the general-purpose multistart optimizer supplied with this report only produces numerical candidates. It is not an exact replacement for quantifier elimination or a verified certificate.

There is additional convex structure for a *fixed* Bob plane. If $K \succ 0$, then $J_n \succ 0$ for every n: the two nonzero eigenvalues of DP_nD^T are those of Q restricted to the plane, so its largest eigenvalue subtracted from k_n leaves at least $\lambda_{\min}(Q) - \|c\|^2 > 0$. For $p < 1$, strict feasibility at $u = a$ gives the exact inner dual

$$\max_{u \in \mathcal{L}_p(a)} f_n(u) = \min_{\lambda,\mu \geq 0} \{ \operatorname{tr}(P_nQ) + \lambda + \mu[(1-p)^2 - r^2] + (h_n + \mu pa)^T [J_n + (\lambda + \mu p^2)I]^{-1}(h_n + \mu pa) \}. \tag{127}$$

This follows by completing the square and the ordinary strictly feasible concave quadratic-program duality theorem. It does not justify interchanging the outer maximum over n with the dual minimum. No universal scalar dual for the original-state frontier is claimed here.

15.5 Universal purity bounds and exact closed families

Let $\alpha = \operatorname{tr}(DD^T) - \|c\|^2$ and $m = (\alpha - 1)/2$. For clarity, the pointwise inequality needed here concerns the *actual effect* u, without any replacement by a reduced Lorentz pair:

$$B((\rho_c)_u)^2 \leq 1 + m(1 - \|u\|^2). \tag{128}$$

One direct proof is as follows. For an arbitrary physical state ω with Bloch data (a', b', T'), a real unit vector n, and $S_n = (n \cdot \sigma) \otimes I$, Pauli orthogonality gives

$$4 \operatorname{tr}(\omega S_n \widetilde{\omega} S_n) = 1 + \|a'\|^2 - \|b'\|^2 - \|T'\|_F^2 - 2(n \cdot a')^2 + 2n^T T' T'^T n \geq 0.$$

The inequality holds because both ω and $S_n \widetilde{\omega} S_n$ are positive semidefinite. Therefore each Alice measurement-plane score satisfies

$$\operatorname{tr}(T'T'^T) - n^T T' T'^T n \leq \frac{1 + \|T'\|_F^2 + \|a'\|^2 - \|b'\|^2}{2} - (n \cdot a')^2.$$

For the actual canonical effect, $a' = u$, $b' = c + D^T u$, and (115) gives

$$\|T'\|_F^2 + \|u\|^2 - \|b'\|^2 = \alpha + (1 - \alpha)\|u\|^2.$$

Taking the maximum over measurement planes and dropping the nonnegative square proves (128), including $\|u\| = 1$. This is the quantitative purity bound from Proposition 2, now written entirely in the coordinates of the constrained problem.

Combining it with the exact norm range of the lens yields

$$S_\rho(p) \leq \begin{cases} 1 + m[1 - r_-(p)^2], & m \geq 0, \\ 1 + m[1 - r_+(p)^2], & m < 0. \end{cases} \tag{129}$$

The following families attain either this bound or a sharper exact spectral expression. Their proofs optimize over all filter directions.

Theorem 27 (Conformal canonical correlations: full-rank and pure frontiers)**.** *Suppose the canonical state is physical and has $c = 0$, $D = tO$, where $t \geq 0$ and O is orthogonal. Then*

$$\boxed{S_\rho(p) = t^2[\,2 - r_-(p)^2\,].} \tag{130}$$

For $t > 0$, the optimal canonical effect is uniquely $u_* = r_-(p)a/r$ when $r > 0$, and $u_* = 0$ when $r = 0$. All these optima are attained by invertible filters. The plateau extends to $p = 1 - r$. This includes arbitrary one-sided invertible-filter images of the Werner/isotropic canonical family, and includes every entangled pure state at $t = 1$.

Proof. Here $G(u) = t^2[(1-\|u\|^2)I+O^T uu^T O]$. Its eigenvalues are t^2, $t^2(1-\|u\|^2)$, and $t^2(1-\|u\|^2)$. The score is therefore $t^2(2 - \|u\|^2)$, so its maximum is reached at the unique minimum-norm point of the lens when $t > 0$. The minimum norm is $r_-(p) \le r < 1$, proving finite attainment. An entangled pure state becomes maximally entangled after whitening, which gives $t = 1$. □

An explicit largest-scale filter on the *original* state is especially simple. In the eigenbasis of ρ_A, with the larger eigenvalue first, use

$$F_p = \begin{cases} \mathrm{diag}\left(\sqrt{\dfrac{1-r}{1+r}}, 1\right), & p \le 1 - r, \\ \mathrm{diag}\left(\sqrt{\dfrac{2p-1+r}{1+r}}, 1\right), & p \ge 1 - r. \end{cases} \tag{131}$$

The first branch succeeds with probability $1 - r$, and the second with probability exactly p. Both are continuous at the breakpoint. For $r = 0$, this prescription is simply the identity.

Corollary 28 (Exact original-input threshold for the full-rank example)**.** *For*

$$\rho_2 = \frac{1}{500}\begin{pmatrix} 441 & 0 & 0 & 56 \\ 0 & 49 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 56 & 0 & 0 & 9 \end{pmatrix},$$

the canonical parameters are $t = 4/5$, $r = 24/25$. Hence

$$S_{\rho_2}(p) = \frac{16}{25}\left[2 - \max\left(0, 1 - \frac{1}{25p}\right)^2\right], \tag{132}$$

with the maximum taken as zero inside the square at $p = 0$. The maximal CHSH score is available up to success $1/25$, whereas some violation is available exactly when

$$\boxed{p < \frac{4(4+\sqrt{7})}{225} \simeq 0.11815.} \tag{133}$$

Proof. The first statement follows from (130). Above the plateau, violation is $1 - 1/(25p) < \sqrt{2 - 25/16} = \sqrt{7}/4$. Solving and rationalizing its positive denominator yields (133). At equality the squared score is one. □

Theorem 29 (Arbitrary maximally correlated rank-two states)**.** *For*

$$\rho_{v,z} = v|00\rangle\langle 00| + (1-v)|11\rangle\langle 11| + z|00\rangle\langle 11| + \overline{z}|11\rangle\langle 00|, \quad 0 < v < 1, \quad |z|^2 \le v(1-v),$$

put $r = |2v - 1|$ and $t^2 = |z|^2/[v(1 - v)]$. Then

$$\boxed{S_{\rho_{v,z}}(p) = 1 + t^2[1 - r_-(p)^2].} \tag{134}$$

The original-state filter (131) attains the optimum. For $z \neq 0$ its positive part is unique up to scale.

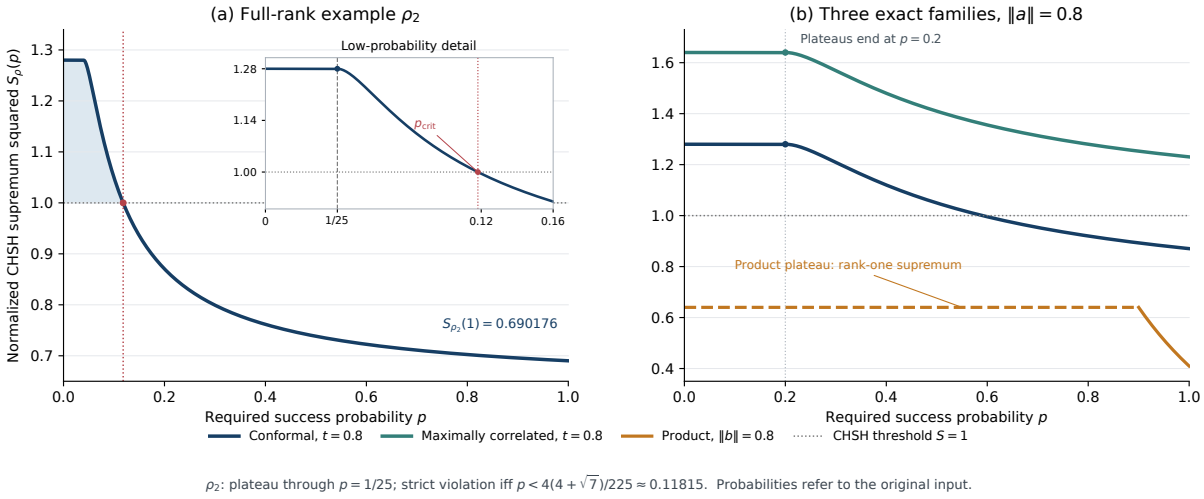
Proof. A phase rotation makes z nonnegative. Whitening gives $c = 0$ and, with the perfect-correlation direction chosen as the third axis, $D = \mathrm{diag}(t, -t, 1)$. Here $m = t^2$, so (129) is precisely the claimed upper bound. The vector a is on the perfect-correlation axis. At the closest feasible effect $u = (0, 0, \pm r_-(p))$, the Gram eigenvalues are $1, t^2(1-r_-(p)^2), t^2(1-r_-(p)^2)$, giving equality. The filter in (131) realizes this effect. If $t > 0$, equality in the upper bound requires the unique minimum-norm point of the lens, proving uniqueness. □

Theorem 30 (All product states, including pure marginals)**.** *For $\rho = \rho_A \otimes \rho_B$, let $r = \|a\|$ and $b_0 = \|b\|$. Then, including the cases $r = 1$ or $b_0 = 0$,*

$$\boxed{S_\rho(p) = b_0^2\, r_+(p)^2.} \tag{135}$$

If $r < 1$ and $b_0 > 0$, the plateau $p \le (1+r)/2$ is a rank-one supremum and is not attained by an invertible filter. For $p > (1+r)/2$, an invertible filter attains the optimum. If Alice is pure, the value is constantly b_0^2 and the identity attains it.

Proof. For a mixed Alice marginal, whitening gives $D = 0$, $c = b$. Equation (115) becomes $G(u) = \|u\|^2\, bb^T$, so the score is $b_0^2\, \|u\|^2$. The largest feasible norm is $r_+(p)$. For $p \le (1+r)/2$ this norm equals one; if $b_0 > 0$, finite attainment would require a pure filtered Alice state, impossible under an invertible filter of a mixed marginal. For larger p, the farthest point is interior, giving finite attainment. Explicitly, preserve the larger marginal eigenvalue $(1+r)/2$ and attenuate the smaller one until the total success is p. If Alice is already pure, every nonzero filter leaves her pure and leaves Bob unchanged, proving the boundary case directly. □



ρ_2: plateau through $p = 1/25$; strict violation iff $p < 4(4+\sqrt{7})/225 \approx 0.11815$. Probabilities refer to the original input.

Figure 4: Exact success frontiers on the original input. The full-rank ρ_2 example violates exactly for required probabilities $p < 4(4+\sqrt{7})/225 \simeq 0.11814669$ and attains its unconstrained score for $p \le 1/25$. The family comparison uses $\|a\| = 0.8$ and $t = 0.8$ for the conformal and maximally correlated states, and $\|b\| = 0.8$ for the product state. The product plateau is a rank-one supremum; the two entangled-family curves have finite attaining filters.

Numerical scope. The file `calculations/original_input/original_success.py` evaluates the exact Bob Gram matrix, projects candidates into the true lens, and performs multistart optimization over all three effect coordinates. Its general output is explicitly labeled as a numerical candidate, not a global certificate. The closed families above and the counterexample are separately checked using direct complex density matrices and exact symbolic identities.

16 Convexity, noise scaling, and quantitative stability

Proposition 31 (Convex sublevel sets and a genuine discontinuity)**.** *For every $b_0 \geq 0$, the set $\{\rho : B_A^*(\rho) \leq b_0\}$ is convex. On each affine slice with a fixed Alice marginal, both B_A^* and $(B_A^*)^2$ are convex functions of ρ. In general B_A^* is neither convex nor continuous at states with pure filtered-side marginal.*

Proof. For a fixed invertible filter, filtering a mixture $\sum_j q_j \rho_j$ produces the mixture of the individually filtered states with weights proportional to their success probabilities. The CHSH norm is convex, because it is the supremum of the appropriate normalized linear Bell expressions. Its value on the filtered mixture is therefore at most $\max_j B_A^*(\rho_j)$. Taking the supremum over filters proves convexity of the sublevel set.

If all ρ_j have the same Alice marginal, each fixed filter has the same success probability on every component. The weights remain q_j, and taking the supremum proves convexity of B_A^* on the slice. The square of a nonnegative convex function is convex, giving the second assertion.

For the last statement, consider

$$\rho_p = p|\Phi^+\rangle\langle\Phi^+| + (1-p)|0\rangle\langle 0| \otimes I/2.$$

At $p = 0$, Alice is pure and Bob maximally mixed, so $B_A^*(\rho_0) = 0$. For $0 < p \leq 1/2$, $\Delta_B(\rho_p) = p(2p-1)/2 \leq 0$, which gives the upper bound one. A rank-one Alice filter onto $|1\rangle$ removes the second component and leaves Bob pure, giving the matching supremum one. Thus $B_A^*(\rho_p) = 1$ throughout $0 < p \leq 1/2$. It jumps at zero and exceeds the convex interpolation $pB_A^*(|\Phi^+\rangle) = p\sqrt{2}$ for, for example, $p = 1/4$. □

Proposition 32 (Exact noise scaling on the unfiltered party)**.** *Let $\mathcal{D}_\eta(X) = \eta X + (1-\eta)\operatorname{tr}(X)I/2$ be the depolarizing channel, with $0 \leq \eta \leq 1$. Then*

$$\boxed{B_A^*((\mathrm{id} \otimes \mathcal{D}_\eta)(\rho)) = \eta B_A^*(\rho).} \tag{136}$$

For $\eta > 0$, the optimal Alice filters are unchanged. If the original optimum exceeds one, the exact surviving-violation threshold is $\eta > 1/B_A^*(\rho)$. For a canonical state the same statement applies to admixture of global white noise.

Proof. Alice filtering commutes with the trace-preserving channel on Bob, and the filtering success probability is unchanged. For every normalized filtered state, the channel multiplies its correlation matrix by η, and hence its CHSH norm by η. The equality follows after taking suprema. If $\rho_A = I/2$, then $(\mathrm{id} \otimes \mathcal{D}_\eta)(\rho) = \eta\rho + (1-\eta)I/4$, proving the last statement. For a noncanonical state, global white noise does not generally obey this scaling law. □

Proposition 33 (A universal canonical Lipschitz estimate)**.** *Let S, S' be the squared unilateral optima of two canonical states, with derived data (A, s, d) and (A', s', d'). Then*

$$\boxed{|S - S'| \leq 2\left\|A - A'\right\|_{\mathrm{op}} + |s - s'| + 2\left\|d - d'\right\|.} \tag{137}$$

This estimate requires no spectral gap and includes all singular and repeated-eigenvalue cases.

Proof. For a common feasible pair, put $u = -n \times v$ and $P = I - nn^T - vv^T$. Because $n \cdot v = 0$, $\|n\| = 1$, and $\|v\| \leq 1$, the matrix P is positive semidefinite and $\operatorname{tr} P = 2 - \|v\|^2 \leq 2$. The objective is

$$f(n, v) = \operatorname{tr}(PA) + s\|v\|^2 + 2u \cdot d.$$

Since $\|u\| = \|v\| \leq 1$, the absolute difference of the two objectives is bounded by the right-hand side of (137), uniformly over the common compact domain. Taking maxima in both orders proves the claim. □

For example, an estimated value $\widehat{S}$ with certified right-hand side at most ε proves activation if $\widehat{S} - \varepsilon > 1$, and excludes activation if $\widehat{S} + \varepsilon \le 1$. The estimate is in canonical coordinates; it does not assert uniform continuity of whitening as a marginal approaches purity.

Proposition 34 (Gradients at a simple optimum)**.** *Suppose $H_0 \succ 0$ and the smallest eigenvalue of M is simple. Choose its normalized eigenvector n, put $v = -K^{-1}Cn$, $u = -n \times v$, and $P = I - nn^T - vv^T$. Then $S = (B_A^*)^2$ is differentiable in the canonical data and*

$$\boxed{\nabla_D S = 2(PD + uc^T), \qquad \nabla_c S = 2(D^T u + \|v\|^2 c).} \tag{138}$$

Proof. Under the stated hypotheses the optimizing reduced pair is unique up to the simultaneous sign change $(n, v) \mapsto (-n, -v)$, which leaves P and u unchanged. The simple-eigenvalue formula is smooth locally. Equivalently, differentiating the maximized objective while holding an optimizing pair fixed gives the same derivative. In $f = \operatorname{tr}(PDD^T) + \|c\|^2 \|v\|^2 + 2u^T Dc$, differentiation with respect to D and c gives precisely (138). □

Independent checks and scope. The accompanying independent script tests these extensions through complex 4×4 density-matrix calculations, separately from the reference solver. With the recorded seed it checks the Gram identity, plane-purity identity, determinant covariance, rational ellipsoid data, and scalar probability formula on 1,000 states of ranks one through four. It tests 417 purity-positive shortcut instances, 600 physical axial states, and 102 success-frontier comparisons against direct multistart filtering. The largest shortcut-attainment discrepancy is 2.7×10^{-15} and the largest frontier discrepancy is 8.2×10^{-10}. Further checks cover 1,000 independent random rank-two states, 129 rational-cubic value evaluations, and 11 points of the exact tilted frontier. The rank-two ceiling discrepancy is below 1.8×10^{-15}; the explicit tilted filters agree with direct optimization within 2.9×10^{-15}. These numerical checks support the mathematical proofs; they do not establish them or extend the Lean coverage. The exact formalization boundary is stated separately in Appendix B.

17 Research process and AI disclosure

This is fully autonomous agentic research. The results, their proofs, the Lean development, the exact and numerical checkers and a draft report were produced by autonomous AI research agents in an earlier session; its records describe internal reviews carried out within the same automated workflow and do not name the system used.

Opus 5.5 by Anthropic used with Claude Code assessed the work for this release. It confirmed that the problem solved is Eq. (9) of [1], read the proofs of the compact reduction, the Gram identity, the least-eigenvalue theorem and the violation criterion, compared the Lean definitions and endpoint statements with the mathematics, ran the fresh Lean build and the whole-project axiom audit recorded in `verification/`, reran every checker from a copy, wrote the independent check `calculations/release_check.py`, read the Omega Institute's dossier on Conjecture 1, and edited and reorganized this manuscript from the draft report, adding the introduction, the main-results summary and the priority notes. The mathematical statements of the draft were not changed. The remaining proofs were carried over and read, but not all were rederived line by line.

Dev D. Goyal is responsible for publication. The work has not been externally peer reviewed; AI checks are not human peer review, no expert validation is claimed, and the literature comparison is bounded and does not establish priority. Appendix B states the formal scope and its limits.

A Proof of the finite-candidate theorem

Section 4 reduced the problem to the maximum (20) and showed that the boundary maximum is E and that interior maxima satisfy the pencil equations (24). It remains to analyse the pencil.

A.1 Why the pencil is a cubic with real roots

Both $\mathcal{H}$ and $\mathcal{B}$ annihilate $\mathcal{L}_0$ and preserve its orthogonal complement. Their restrictions have no common kernel. If $(\widehat{H} - i\widehat{B})z = 0$, its Hermitian inner product with z has imaginary part $-z^*\widehat{B}z = 0$. Positive semidefiniteness of $\widehat{B}$ implies $\widehat{B}z = 0$ and then $\widehat{H}z = 0$, a contradiction. Thus $p(i) \neq 0$, so p is nonzero. Its degree is at most $\operatorname{rank}\widehat{B} \leq 3$.

If $(\widehat{H} - \lambda\widehat{B})z = 0$ for a nonzero complex z, then $z^*\widehat{B}z > 0$ by the same common-kernel argument. Hence

$$\lambda = \frac{z^*\widehat{H}z}{z^*\widehat{B}z} \in \mathbb{R}. \tag{139}$$

Moreover, projecting an eigenspace representative onto its first three coordinates is injective; a representative with $n = 0$ would lie in the removed common kernel. Thus every reduced eigenspace has dimension at most three.

A.2 The feasibility test and the exceptional level

For $\lambda \neq s$, (25) already forces $n^T v = 0$. A nonzero solution can be normalized to $\|n\| = 1$, $\|v\| \leq 1$ exactly when some solution has

$$\|n\|^2 - \|v\|^2 \geq 0. \tag{140}$$

Adding a common-kernel vector changes only the component of v in $\ker K \cap \ker C$. Removing that component maximizes the displayed difference. The existence test is therefore exactly $\chi_\lambda \geq 0$, and its strict version gives $\|v\| < 1$.

At $\lambda = s$, the equations are the real and imaginary parts of

$$(K + iC)(n + iv) = 0. \tag{141}$$

They are invariant under multiplication of $n + iv$ by a complex phase. Choose that phase so that the real and imaginary parts are orthogonal and the real part has at least the imaginary part's norm. This is just diagonalization of the real 2×2 Gram matrix of n, v. Then normalize the real part to norm one. The resulting pair belongs to $\mathcal{D}$ and has the required value. If $\chi_s > 0$, the phase can give strict inequality of norms. Conversely, an interior representative has a positive $\mathcal{J}$-quadratic form, so it implies $\chi_s > 0$ after removing the common-kernel component.

An equivalent exact strict test is useful in implementations: for any complex basis matrix Q_c of $\ker(K + iC)$, the exceptional level has an interior representative if and only if

$$Q_c^T Q_c \neq 0. \tag{142}$$

Indeed $z^T z \neq 0$ means that a phase can make $z^T z$ positive real, giving orthogonal real and imaginary parts with unequal norms. If all these bilinear squares vanish, every phase-normalized representative is on the boundary. This handles repeated eigenvalues without choosing an arbitrary real eigenvector.

Every admissible spectral level therefore has a feasible representative, while every interior maximum is among those levels. Together with the boundary maximum, this proves Theorem 6.

A.3 Justification of the singular compression

From $Kv = -Cn$ we obtain $n \perp C \ker K$, so $n \in \operatorname{ran} P$. The general solution is $v = -K^{\dagger}Cn + Nw$. Substitution into the first stationary equation yields

$$(M_0 - \lambda I)n = CNw. \tag{143}$$

Projecting to $(C \ker K)^{\perp}$ gives exactly the eigenproblem for M_L. The residual lies in $\operatorname{ran}(CN)$, and the minimum-norm solution for w gives (57). All remaining freedom is a vector in $\ker K \cap \ker C$, which only increases $\|v\|$ from its minimum. For $\lambda \neq s$, orthogonality to n is automatic for every such solution. This proves the stated singular-value test and demonstrates that no regularization limit is hidden in the formula.

B Verification and formal scope

B.1 What the Lean development proves

The Lean project in `lean/` pins Lean 4.19.0 and Mathlib revision `c44e0c8ee63ca166450922a373c7409c5d26b00b`. Its 32 proof modules use the following definitions. A state is a matrix $\rho \in \mathbb{C}^{4\times 4}$, indexed by pairs of qubit indices, with the hypotheses that ρ is positive semidefinite (Mathlib's `PosSemidef`) and has trace one. An Alice filter is any $F \in \mathbb{C}^{2\times 2}$ with $\det F \neq 0$; the filtered state is $(F \otimes I)\rho(F \otimes I)^{\dagger}$ divided by the real part of its trace, and the development proves that this trace is positive and that the result is again a density matrix. Bob filters act on the second factor. A measurement observable is a 2×2 complex matrix O with $O^{\dagger} = O$, $O^2 = I$ and $\operatorname{tr} O = 0$; this is proved equivalent to $O = \hat{u} \cdot \sigma$ for a unit vector $\hat{u}$. The value attached to a filter and four observables is the square of $\Re \operatorname{tr}(\rho' \mathcal{B})/2$, where $\mathcal{B}$ is the usual Bell operator, so that the local bound is 1; the measure is the supremum (least upper bound) of these values. As in [1], filters are not required to be contractions: a scalar multiple of a filter gives the same normalized state.

Statement	Lean declaration (namespace `UnilateralCHSH.Quantum`)
Result A: both one-sided suprema, every state, arbitrary observables	`all_states_operator_both_isLUB`
Pure filtered marginal: value $\|b\|^2$, attained	`pure_marginal_gl_isGreatest`
Result B: least-eigenvalue formula for $\alpha > 1$ with an attaining filter	`all_states_alice_schur_formula`
Result C: violation criterion (Alice, Bob)	`all_states_alice_operator_activation_iff`, `all_states_bob_operator_activation_iff`
Hidden one-sided activation	`all_states_alice_operator_hidden_activation_iff` and the Bob version
Invertible attainment of a violating supremum	`all_states_alice_operator_violation_attained` and the Bob version
Result D: at most four candidates (boundary value and spectral levels)	`original_finite_candidates`
Whitening in explicit coordinates, orbit equality	`original_filter_orbit_explicit`
Bob's measure from Alice's by an actual qubit swap	`bobGLValues_eq_alice_swap`

In these statements the value of the supremum is the function `aliceMaximum`: for a mixed Alice marginal it is the maximum of the reduced objective (15) over the compact domain (16), evaluated at the whitened coordinates, and for a pure marginal it is $\|b\|^2$. The least-eigenvalue statement asserts $\det K \neq 0$ and produces $r > 0$ and a unit vector n with $(K + CK^{-1}C)n = rn$,

$r\,\|x\|^2 \le x^{\mathsf{T}}(K + CK^{-1}C)x$ for every x, value $\alpha - r$, and an invertible filter attaining it; with $M - sI = K + CK^{-1}C$ this is Result B. The violation criterion is stated as $\|a\| < 1$, $\alpha > 1$ and the existence of a unit vector n with $n^{\mathsf{T}}(K + CK^{-1}C)n < \alpha - 1$, which is $\lambda_{\min}(M) < \operatorname{tr} A - 1$. Along the way the development proves the compact attainment and Lagrange conditions derived from actual maximality, the finite spectral candidates including all singular cases, the reduction of every filter and measurement plane, the Gram identity and purity guard, the Horodecki plane formula with explicit attaining measurements, local-unitary invariance, polar factorization of every invertible filter, the Fano representation, whitening and its inverse, continuity and radial approximation of the supremum, and product factorization at a pure marginal.

Audit. The audit module `UnilateralAxiomAudit` enumerates every declaration whose defining module is one of the 32 proof modules and collects its axioms with Lean's `collectAxioms`. In the fresh build recorded in `verification/` it covers 1,060 declarations. The 869 declarations other than code-generation auxiliaries are not axioms and depend only on `propext`, `Classical.choice` and `Quot.sound`; the other 191 are auxiliary declarations that the Lean 4.19 code generator stores for executable definitions, 12 of which refer to its placeholder `lcProof`, which the audit permits for such auxiliaries only. All 730 named source declarations were audited, and the sources contain no `sorry`, `admit`, `native_decide`, `ofReduceBool`, `implemented_by`, `extern`, `unsafe` or `axiom`.

B.2 What is conventional

The following have conventional proofs in this paper, with the exact and numerical checks listed below, but are not formalized: the explicit evaluation of the boundary radius E by the secular equation and its hard case (Section 8), the explicit cubic, the admissibility tests and the singular compression of Section 7 (the formal finite-candidate theorem uses an attained maximum over the sphere and the eligible spectral levels); the polynomial and rational-ellipsoid activation tests and the rational cubic (Section 10); the three regimes at $B_A^* = 1$ and the product-kernel criterion; the rank-two formula, which uses the Ruskai–Szarek–Werner classification [11]; the characterization of $B_A^* = \sqrt{2}$; the axial family and the quasi-distillable family; all success-probability results of Sections 14 and 15; and the convexity, scaling and stability results of Section 16. The extension of Result B to the hypothesis $K + iC \succ 0$ is also conventional.

B.3 Exact and numerical checks

The directory `calculations/` contains the checkers and their recorded outputs; `scripts/run_checks.py` replays all of them in a temporary copy with `python3 -I` and compares each output with the record. They are falsification attempts and exact checks of particular instances; no theorem rests on them.

- `exact_examples.py`: exact SymPy verification of positivity, Bloch data, candidate eigenvalues, filters, success probabilities and output correlations for the examples of Section 9 and the following sections, and of the symbolic pencil determinant identity.
- `audit_spectral_cases.py`: 655 canonical data sets, including $\operatorname{rank} K \in \{0, 1, 2\}$, common kernels, repeated eigenvalues and vanishing centres, with an independent spectral reconstruction and 252,500 sampled filters; reconstructed witnesses match to about 10^{-13} and no sampled filter exceeds the formula.
- `independent_audit/check_spectral.py`: 275 comparisons of the formula with multistart direct filtering of complex density matrices (canonical states of ranks one to four, original states for both parties, singular cases, exact examples); direct optimization never exceeds

the formula by more than roundoff, with a largest gap of about 4.4×10^{-8} consistent with approaching a boundary supremum.

- `extensions/independent_checks.py`: 1,000 states of ranks one to four for the Gram, plane-purity, covariance and ellipsoid identities; 417 guard-region instances; 600 axial states; 102 success-frontier comparisons with direct filtering; the counterexample showing that $K \succ 0$ alone does not suffice.

- `extensions/additional_exact_checks.py`, `rank2_checks.py`, `nonunique_optimizer_example.py`, `bilateral_family_certificate.py`: the rational cubic on sampled states, 1,000 rank-two states, an exact circle of optimal filters, and an exact certificate for the bilateral filter of the quasi-distillable family at $p = 3/5$.

- `original_input/`: 256 original states with 512 arbitrary complex filter comparisons for the whitening formulas, and the original-input success lens, Gram matrix and closed frontiers.

- `check_extensions.py`: tests of the public solver functions, including exact activation signs and exact cubic roots.

- `check_filter_reduction.py`: 2,000 instances of the real filter reduction used in the formal proof.

- `release_check.py`, written for this release without importing the other files: exact values for ρ_1, ρ_2 and the quasi-distillable family on the grid $p = k/12$, the explicit optimal filter of ρ_1 applied to the density matrix, and 160 random states of ranks one to four with $\alpha > 1$, for which the least-eigenvalue value agrees with the reconstructed invertible filter and with multistart direct maximization to about 10^{-13}.

Numerical rank caution. The reference solver uses floating-point eigensolvers and tolerances and is not an exact certificate near degeneracies. There is a genuine discontinuity at a pure filtered-side marginal: for $\sqrt{1-\varepsilon}|00\rangle + \sqrt{\varepsilon}|11\rangle$, every $\varepsilon > 0$ has $B_A^* = \sqrt{2}$, whereas $\varepsilon = 0$ has value 1. Treating a small positive marginal eigenvalue as zero can therefore produce an order-one error; the solver refuses unsafe near-singular marginal decisions instead of returning a product-state answer.

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